

Asymptotic Approximation Methods

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1 Introduction

This course is on methods for solving asymptotic problems, i.e. problems involving a small parameter. We'll look at 3 methods:

1. Matched asymptotic expansions
2. Method of multiple scales
3. WKB approximation

2 Matched asymptotic expansions

2.1 Notations and definitions

Consider continuously differentiable functions $f(\varepsilon)$ and $\phi(\varepsilon)$ defined on an interval $\varepsilon_0 < \varepsilon < \varepsilon_1$.

Definition 2.1. (Big O notation) We say that $f(\varepsilon) = O(\phi(\varepsilon))$ as $\varepsilon \downarrow \varepsilon_0$ if there exists $K > 0$ and $\varepsilon_* > \varepsilon_0$ such that $|f(\varepsilon)| < K|\phi(\varepsilon)|$ for all $\varepsilon \in (\varepsilon_0, \varepsilon_*)$

Definition 2.2. (Little o notation) We say $f(\varepsilon) = o(\phi(\varepsilon))$ as $\varepsilon \downarrow \varepsilon_0$ if for any given $\delta > 0$ there exists $\varepsilon_* > \varepsilon_0$ such that

$$|f(\varepsilon)| < \delta|\phi(\varepsilon)| \text{ for all } \varepsilon \in (\varepsilon_0, \varepsilon_*)$$

this is often written as $f(\varepsilon) \ll \phi(\varepsilon)$ as $\varepsilon \downarrow \varepsilon_0$.

Theorem 2.3. (Mini-theorems)

(i)

$$\lim_{\varepsilon \downarrow \varepsilon_0} \frac{f(\varepsilon)}{\phi(\varepsilon)} = K \implies f(\varepsilon) = O(\phi(\varepsilon))$$

(ii)

$$\lim_{\varepsilon \downarrow \varepsilon_0} \frac{f(\varepsilon)}{\phi(\varepsilon)} = 0 \implies f(\varepsilon) = o(\phi(\varepsilon))$$

Example. Show that $f(\varepsilon) = -2\varepsilon^3 + 3\varepsilon^5$ is $O(\varepsilon^3)$ as $\varepsilon \downarrow 0$.

Proof. Using the Mini-theorems(i) we see that

$$\lim_{\varepsilon \downarrow 0} \frac{-f(\varepsilon)}{\varepsilon^3} = 2 \implies -f(\varepsilon) = O(\varepsilon^3)$$
$$\implies |-2f(\varepsilon)| < 2|\varepsilon| \text{ as } \varepsilon \downarrow 0$$

i.e. $|f(\varepsilon)| < 2|\varepsilon^3|$ as $\varepsilon \downarrow 0 \implies f(\varepsilon) = O(\varepsilon^3)$ as $\varepsilon \downarrow 0$. □

Example. Show that $f(\varepsilon) = \varepsilon \sin(1 + \frac{1}{\varepsilon})$ is $O(\varepsilon)$ as $\varepsilon \downarrow 0$.

Proof. Note, Mini-theorems is not helpful here since

$$\lim_{\varepsilon \downarrow 0} \frac{f(\varepsilon)}{\varepsilon} = \lim_{\varepsilon \downarrow 0} \sin\left(1 + \frac{1}{\varepsilon}\right)$$

which does not exist. Instead, go back to the definition of O . Note that

$$|f(\varepsilon)| = |\varepsilon| \left| \sin\left(1 + \frac{1}{\varepsilon}\right) \right| \leq |\varepsilon| < 2|\varepsilon|$$

thus $f(\varepsilon) = \varepsilon \sin(1 + \frac{1}{\varepsilon})$ is $O(\varepsilon)$ as $\varepsilon \downarrow 0$. □

Definition 2.4. The set of functions $\{\phi_m(\varepsilon) : m \in [1, n] \cap \mathbb{Z}\}$ form an asymptotic sequence as $\varepsilon \downarrow \varepsilon_0$ if $\phi_{m+1} = o(\phi_m)$ as $\varepsilon \downarrow \varepsilon_0$ for $1 \leq m \leq n-1$.

Example. $\phi_m(\varepsilon) = \varepsilon^m$ is an asymptotic sequence as $\varepsilon \downarrow 0$.

Proof. By Mini-theorems(ii),

$$\lim_{\varepsilon \downarrow 0} \frac{\varepsilon^{m+1}}{\varepsilon^m} = \lim_{\varepsilon \downarrow 0} \varepsilon = 0$$

this is indeed an asymptotic sequence. □

Example. $\phi_m(\varepsilon) = (\varepsilon - \varepsilon_0)^{\alpha_m}$ where $\alpha_1 < \alpha_2 < \dots$ is an asymptotic sequence as $\varepsilon \downarrow 0$.

Proof. By Mini-theorems(ii),

$$\lim_{\varepsilon \downarrow \varepsilon_0} \frac{(\varepsilon - \varepsilon_0)^{\alpha_{m+1}}}{(\varepsilon - \varepsilon_0)^{\alpha_m}} = \lim_{\varepsilon \downarrow 0} (\varepsilon - \varepsilon_0)^{\alpha_{m+1} - \alpha_m} = 0$$

this is indeed an asymptotic sequence. $\square$

Example. $\phi_1(\varepsilon) = 1, \phi_2(\varepsilon) = \varepsilon \log \varepsilon, \phi_3(\varepsilon) = \varepsilon, \phi_4(\varepsilon) = \varepsilon^2 \log \varepsilon, \phi_5(\varepsilon) = \varepsilon^2, \dots$ is an asymptotic sequence as $\varepsilon \downarrow 0$.

Proof. By Mini-theorems(ii),

$$\lim_{\varepsilon \downarrow 0} \frac{\phi_{m+1}}{\phi_m} = \begin{cases} \lim_{\varepsilon \downarrow 0} \frac{1}{\log \varepsilon} = 0 & \text{for } m \text{ even} \\ \lim_{\varepsilon \downarrow 0} \varepsilon \log \varepsilon = 0 & \text{for } m \text{ odd} \end{cases}$$

this is indeed an asymptotic sequence. $\square$

Example. $\phi_m(\varepsilon) = e^{-(m-1)/\varepsilon}$ is an asymptotic sequence as $\varepsilon \downarrow 0$.

Proof. By Mini-theorems(ii),

$$\lim_{\varepsilon \downarrow 0} \frac{e^{-m/\varepsilon}}{e^{-(m-1)/\varepsilon}} = \lim_{\varepsilon \downarrow 0} e^{-1/\varepsilon} = 0$$

this is indeed an asymptotic sequence. $\square$

Definition 2.5. We say that a function $f(\varepsilon)$ is transcendentally small or asymptotically small beyond all orders as $\varepsilon \downarrow \varepsilon_0$ if $f(\varepsilon) = o((\varepsilon - \varepsilon_0)^p)$ for all $p > 0$ as $\varepsilon \downarrow \varepsilon_0$.

Example. For any $p > 0$,

$$\lim_{\varepsilon \downarrow 0} \frac{e^{-1/\varepsilon}}{\varepsilon^p} = \lim_{x \rightarrow \infty} \frac{e^{-x}}{(\frac{1}{x})^p} = \lim_{x \rightarrow \infty} x^p e^{-x} = 0$$

where substitution $x = \frac{1}{\varepsilon}$ is used. By Mini-theorems(ii), $e^{-1/\varepsilon} = o(\varepsilon^p)$ as $\varepsilon \downarrow 0$ for any $p > 0$. i.e. $e^{-1/\varepsilon}$ is transcendentally small as $\varepsilon \downarrow 0$.

Definition 2.6. A function $f(\varepsilon)$ is said to have an asymptotic expansion to order n in the asymptotic sequence $\{\phi_m(\varepsilon) : m \in [1, n] \cap \mathbb{Z}\}$, if there exists constant $\{a_m : m \in [1, n] \cap \mathbb{Z}\}$ such that

$$f(\varepsilon) = \sum_{k=1}^m a_k \phi_k(\varepsilon) + o(\phi_m)$$

for all $m \in [1, n] \cap \mathbb{Z}$.

Example. If $f(\varepsilon) \in C^\infty$ at $\varepsilon = 0$ we know that it has a Taylor series expansion in powers of ε .

Remark. These constants in an asymptotic expansion are unique. i.e., if we have

$$\begin{aligned} f(\varepsilon) &= \sum_{k=1}^m a_k \phi_k(\varepsilon) + o(\phi_m(\varepsilon)) \\ &= \sum_{k=1}^m b_k \phi_k(\varepsilon) + o(\phi_m(\varepsilon)) \end{aligned}$$

then $a_k = b_k$ for all k .

2.2 Regular and singular perturbations for the roots of a polynomial

Asymptotic problems can be classified as one of the following:

1. Regular problems: These are ones for which the solutions as $\varepsilon \rightarrow 0$ is the same as the $\varepsilon = 0$ solution.
2. Singular problems: These are ones for which the $\varepsilon = 0$ solution differs in character in a fundamental way from the solution we get as $\varepsilon \rightarrow 0$.

Example. (Regular): Seek the roots of the polynomial

$$x^2 - (3 + \varepsilon)x + 2 = 0$$

We could write down these roots using the exact formula for the roots of a quadratic. This would give

$$x_{\pm} = \frac{(3 + \varepsilon) \pm \sqrt{(3 + \varepsilon)^2 - 8}}{2}$$

and we could then expand the square using the general binomial formula:

$$\sqrt{1 + 6\varepsilon + \varepsilon^2} = 1 + \frac{1}{2}(6\varepsilon + \varepsilon^2) + \frac{\frac{1}{2}-1}{2!}(6\varepsilon + \varepsilon^2)^2 + \dots$$

But instead, let us begin by seeking these roots in the form

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

for some $x_0, x_1, \dots$ to be determined. Substitute this form into our polynomial gives:

$$\begin{aligned}(x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots)^2 - (3 + \varepsilon)(x_0 + \varepsilon x_1 + \varepsilon^2 x_2) + 2 &= 0 \\ \implies x_0^2 - 3x_0 + 2 &= 0 \text{ at } O(1) \\ \implies 2x_0x_1 - 3x_1 - x_0 &= 0 \text{ at } O(\varepsilon) \\ \implies 2x_0x_2 + x_1^2 - 3x_2 - x_1 &= 0 \text{ at } O(\varepsilon^2)\end{aligned}$$

thus, this means

$$\begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix} \text{ or } \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

thus, one root is $x_- = 1 - \varepsilon + 2\varepsilon^2 + O(\varepsilon^3)$ and another root is $x_+ = 2 + 2\varepsilon - 2\varepsilon^2 + O(\varepsilon^3)$.

Example. (Singular): Find the solutions of

$$\varepsilon x^3 + x^2 - 3x + 2 = 0$$

As above, seek a solution of the form

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + O(\varepsilon^3)$$

for some $x_0, x_1, x_2, \dots$ to be determined. Substitute this into the equation:

$$\varepsilon(x_0 + \varepsilon x_1 + \dots)^3 + (x_0 + \varepsilon x_1 + \dots)^2 - 3(x_0 + \varepsilon x_1 + \dots) + 2 = 0$$

Now set coefficient of ε^n to 0 for $n \geq 0$. At $O(1)$,

$$x_0^2 - 3x_0 + 2 = 0$$

so $x_0 = 1$ or 2 . At $O(\varepsilon)$,

$$\begin{aligned}x_0^3 + 2x_0x_1 - 3x_1 &= 0 \\ \implies x_1 &= \frac{x_0^3}{3 - 2x_0}\end{aligned}$$

so $x_1 = 1$ or -8 . Thus, $x = 1 + \varepsilon + O(\varepsilon^2)$ and $x = 2 - 8\varepsilon + O(\varepsilon^2)$. To find the third solution our original equation is

$$\varepsilon x^3 + x^2 - 3x + 2 = 0$$

The only possibility is that this third solution of the form $x = O(\frac{1}{\delta(\varepsilon)})$ for some $\delta(\varepsilon)$ which tend to zero for $\varepsilon \rightarrow 0$. To find the third solution, introduce $X = \delta(\varepsilon)x$. In terms of X ,

$$\begin{aligned}\varepsilon\left(\frac{X}{\delta(\varepsilon)}\right)^3 + \left(\frac{X}{\delta(\varepsilon)}\right)^2 - 3\left(\frac{X}{\delta(\varepsilon)}\right) + 2 &= 0 \\ \implies \varepsilon X^3 + \delta X^2 - 3\delta^2 X + 2\delta^3 &= 0\end{aligned}$$

We will seek solutions for X of the form

$$X = X_0 + \varepsilon X_1 + \varepsilon^2 X_2 + O(\varepsilon^3)$$

where $X_0 \neq 0$. Try $\delta = \varepsilon$. Then

$$X^3 + X^2 - 3\varepsilon X + 2\varepsilon^2 = 0$$

Substitute X into the equation above

$$(X_0 + \varepsilon X_1 + \dots)^3 + (X_0 + \varepsilon X_1 + \dots)^2 - 3\varepsilon(X_0 + \varepsilon X_1 + \dots) + 2\varepsilon^2 = 0$$

At $O(1)$,

$$X_0^3 + X_0^2 = 0$$

so $X_0 = 0$ or -1 . Dismiss $X_0 = 0$ At $O(\varepsilon)$,

$$\begin{aligned}3X_0^2X_1 + 2X_0X_1 - 3X_0 &= 0 \\ \implies 3X_1 - 2X_1 + 3 &= 0 \implies X_1 = 3 \\ \implies X &= -1 - 3\varepsilon + O(\varepsilon^2)\end{aligned}$$

thus the third root is

$$x = -\frac{1}{\varepsilon} - 3 + O(\varepsilon)$$

Example. Consider the equation

$$\varepsilon^7 x^5 - 3\varepsilon^4 x^4 + 2\varepsilon x^3 + x^2 - 3x + 2 = 0$$

If we seek solutions of the form $x = x_0 + \varepsilon x_1 + O(\varepsilon^2)$ we will find only two solutions, namely $x = 1 + O(\varepsilon)$ and $x = 2 + O(\varepsilon)$. To find the remaining solutions introduction $X = \delta(\varepsilon)x$ for some $\delta(\varepsilon)$ to be chosen appropriately. Then our equation in terms of X is

$$\begin{aligned}\frac{\varepsilon^7 X^5}{\delta^5} - \frac{3\varepsilon^4 X^4}{\delta^4} + \frac{2\varepsilon X^3}{\delta^3} + \frac{X^2}{\delta^2} - \frac{3X}{\delta} + 2 &= 0 \\ \implies \varepsilon^7 X^5 - 3\varepsilon^4 \delta X^4 + 2\varepsilon \delta^2 X^3 + \delta^3 X^2 - 3\delta^4 X + 2\delta^5 &= 0\end{aligned}$$

Try $\delta = \varepsilon$ then

$$\begin{aligned}\varepsilon^2 X^5 - 3X^4 + \frac{2X^3}{\varepsilon^2} + \frac{X^2}{\varepsilon^2} - \frac{3X}{\varepsilon} + 2 &= 0 \\ \implies \varepsilon^4 X^5 - 3\varepsilon^2 X^4 + 2X^3 + X^2 - 3\varepsilon X + 2\varepsilon^2 &= 0\end{aligned}$$

by substitution $X = X_0 + \varepsilon X_1 + \varepsilon^2 X_2 + \dots$ we have

$$2X_0^3 + X_0^2 = 0 \implies X_0 = -\frac{1}{2}$$

so $X = -\frac{1}{2} + O(\varepsilon) \implies x = -\frac{1}{2\varepsilon} + O(1)$. Next try $\delta = \varepsilon^2$ then

$$\begin{aligned} \frac{X^5}{\varepsilon^3} - \frac{3X^4}{\varepsilon^4} + \frac{2X^3}{\varepsilon^5} + \frac{X^2}{\varepsilon^4} - \frac{3X}{\varepsilon^2} + 2 &= 0 \\ \implies \varepsilon^2 X^5 - 3\varepsilon X^4 + 2X^3 + \varepsilon X^2 - 3\varepsilon^3 + 2\varepsilon^5 &= 0 \end{aligned}$$

then $X_0 = 0$ so this does not give any new solutions. Next, try $\delta(\varepsilon) = \varepsilon^p$ then

$$\begin{aligned} \frac{\varepsilon^7 X^5}{\varepsilon^{5p}} - \frac{3\varepsilon^4 X^4}{\varepsilon^{4p}} + \frac{2\varepsilon X^3}{\varepsilon^{3p}} + \frac{X^2}{\varepsilon^{2p}} - \frac{3X}{\varepsilon^p} + 2 &= 0 \\ \implies 0 = 2\varepsilon^{5p} + \varepsilon^7 X^5 - 3\varepsilon^{4+p} X^4 + 2\varepsilon^{1+2p} X^3 + \varepsilon^{3p} X^2 - 3\varepsilon^{4p} X & \end{aligned}$$

Once again, seek X of the form $X = X_0 + \varepsilon X_1 + \varepsilon^2 X_2 + \dots$. Suppose $7 = 4 + p$ so $p = 3$ then at leading order,

$$X_0^5 - 3X_0^4 + 2X_0^3 = 0$$

thus more than two terms have survived, i.e., we have "leading order balance".

$$\implies X_0 = 1 \text{ or } 2$$

$$\implies X = 1 + O(\varepsilon), 2 + O(\varepsilon)$$

$$\implies x = \frac{1}{\varepsilon^3} + O(\varepsilon^{-2}), \frac{2}{\varepsilon^3} + O(\varepsilon^{-2})$$

are the remaining two solutions of our original equation.

2.3 Boundary value problems

There are some similarities between the polynomial problems we have just seen and boundary value problems involving some small parameter ε .

Example. Consider the linear differential equation (with boundary conditions)

$$y'' - (3 + \varepsilon)y' + 2y = 0$$

is a regular problem. However,

$$\varepsilon y''' + y'' - 3y' + 2y = 0$$

is a singular problem. Also note that the characteristic equation of this ODE is the polynomial equation

$$\varepsilon m^3 + m^2 - 3m + 2 = 0$$

which is also a singular problem. This has a solution for m which behaves like $\frac{1}{\delta(\varepsilon)}$ where $\delta(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. The corresponding solution for $y = Ae^{mx}$ will therefore grow or decay exponentially fast as $\varepsilon \rightarrow 0$. This gives rise to what is known as a "boundary layer".

Example. (A regular ODE) For $0 \ll \varepsilon \ll \varepsilon$

$$y'' + \varepsilon(y')^3 + y = 0$$

on $0 < x < \frac{\pi}{2}$ with $y(0) = 0$ and $y(\frac{\pi}{2}) = 1$. Seek solutions for $y(x)$ of the form

$$y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$$

Substituting this expansion into the ODE gives

$$\begin{aligned} (y_0'' + \varepsilon y_1'' + \dots) + \varepsilon(y_0' + \dots)^3 + (y_0 + \varepsilon y_1 + \dots) &= 0 \\ \implies (y_0'' + y_0) + \varepsilon(y_1'' + (y_0')^3 + y_1) + O(\varepsilon^2) &= 0 \end{aligned}$$

So we have, at $O(1)$,

$$y_0'' + y_0 = 0$$

at $O(\varepsilon)$,

$$y_1'' + (y_0')^3 + y_1 = 0$$

When we look at the boundary conditions,

$$y(0) = 0 \implies y_0(0) + \varepsilon y_1(0) + O(\varepsilon^2) = 0 \implies y_0(0) = 0, y_1(0) = 0$$

Similarly,

$$y(\frac{\pi}{2}) = 1 \implies y_0(\frac{\pi}{2}) + \varepsilon y_1(\frac{\pi}{2}) + O(\varepsilon^2) = 1 \implies y_0(\frac{\pi}{2}) = 1, y_1(\frac{\pi}{2}) = 0$$

So, for y_0 , we have $y_0'' + y_0 = 0$ with $y_0(0) = 0$ and $y_0(\frac{\pi}{2}) = 1$. This has solution $y_0(x) = \sin(x)$. Next, for y_1 ,

$$y_1'' + y_1 = -\cos^3(x)$$

with $y_1(0) = y_1(\frac{\pi}{2}) = 0$. One finds

$$y_1(x) = -\frac{1}{32} \cos(x) + \frac{3\pi}{16} \sin(x) + \frac{1}{32} \cos(3x) - \frac{3x}{8} \sin(x)$$

Example. (A singular ODE) For $0 < \varepsilon \ll 1$, solve

$$\varepsilon y'' + y' + y = 0$$

on $0 < x < 1$ with $y(0; \varepsilon) = 0$ and $y(1; \varepsilon) = e^{-1}$.

We could find the exact solution for $y(x)$ to this boundary value problem. It is

$$y(x; \varepsilon) = e^{-1} \left(\frac{e^{P_+ x} - e^{P_- x}}{e^{P_+} - e^{P_-}} \right)$$

where

$$P_{\pm} = \frac{-1 \pm \sqrt{1 - 4\varepsilon}}{2\varepsilon}$$

This solution for $y(x; \varepsilon)$ changes very rapidly over a short range of x near to $x = 0$. This is known as a boundary layer. We will now seek to reconstruct this solution using an asymptotic expansions. We will seek one expansion for $x = O(1)$ (the outer expansion) and another for x in the boundary layer (the inner expansion). To find the outer expansion, seek $y(x; \varepsilon)$ of the form

$$y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + \dots$$

substituting this into the ODE gives

$$\begin{aligned} \varepsilon(y_0'' + \varepsilon y_1'' + \dots) + (y_0' + \varepsilon y_1' + \dots) + (y_0 + \varepsilon y_1 + \dots) &= 0 \\ \implies (y_0' + y_0) + \varepsilon(y_0'' + y_1' + y_1) + \dots &= 0 \end{aligned}$$

with boundary conditions $y_0(0) = 0$, $y_0(1) = e^{-1}$. $y_1(0) = y_1(1) = 0$. So for $y_0(0)$ we have $y_0' + y_0 = 0$. Since we are seeking an outer expansion for $y(x; \varepsilon)$ impose the condition on $y_0(1)$, i.e., $y_0(1) = e^{-1}$. This gives $y_0(x) = e^{-x}$. Next, consider $y_1(x)$ for this

$$y_1' + y_1 = -y_0'' = -e^{-x}$$

with the boundary condition $y_1(1) = 0$ then $y_1(x) = (1 - x)e^{-x}$. So the outer expansion for $y(x; \varepsilon)$ is given by

$$y(x; \varepsilon) = e^{-x} + \varepsilon(1 - x)e^{-x} + \dots$$

this approximates $y(x; \varepsilon)$ well for $x = O(1)$ but not for very small x . Let us now seek an expansion for $y(x; \varepsilon)$ that does approximate $y(x; \varepsilon)$ well for small x . This is known as an inner expansion. Let us consider $x = O(\varepsilon)$, introduce a new variable X given by $x = \varepsilon X$ where $X = O(1)$, in terms of X our ODE becomes

$$\frac{d^2 y}{dX^2} + \frac{dy}{dX} + \varepsilon y = 0$$

since

$$\frac{d}{dx} = \frac{d}{dX} \frac{dX}{dx} = \frac{1}{\varepsilon} \frac{d}{dX}$$

Now seek $y(X)$ in the form

$$y = y_0(X) + \varepsilon y_1(X) + O(\varepsilon^2)$$

Substituting this expansion into the ODE gives

$$(y_0'' + \varepsilon y_1'' + \dots) + (y_0' + \varepsilon y_1' + \dots) + \varepsilon(y_0 + \varepsilon y_1 + \dots) = 0$$

Our boundary condition $y(0; \varepsilon) = 0 \implies y_0(0) = 0, y_1(0) = 0$. At $O(1)$ we have

$$\begin{aligned} y_0'' + y_0' &= 0 \\ \implies y_0 &= A_0(1 - e^{-X}) \end{aligned}$$

At $O(\varepsilon)$ we have

$$\begin{aligned} y_1'' + y_1' + y_0 &= 0 \\ \implies y_1(X) &= -A_0 X - A_0 X e^{-X} + A_1(1 - e^{-X}) \end{aligned}$$

where A_1 is another constant. So our inner expansion is (up to $O(\varepsilon)$)

$$y = A_0(1 - e^{-X}) + \varepsilon(-A_0 X - A_0 X e^{-X} + A_1(1 - e^{-X})) + O(\varepsilon^2)$$

Let us now attempt to match our inner and outer expansions for $y(x; \varepsilon)$. This will determine the constants A_0 and $A_1, \dots$. Our outer expansion is valid for $x = O(1)$ while our inner expansion is valid for $x = O(\varepsilon)$. However, it is reasonable to assume there should be some overlapping region where both are valid. There are various strategies for matching expansions. We will use Prandtl's matching principle: Prandtl proposed matching expansions according to the rule

$$\lim_{x \rightarrow 0} y_{\text{outer}} = \lim_{X \rightarrow \infty} y_{\text{inner}}$$

However, this only works for matching leading order expansions. For our example, it gives the following

$$\lim_{x \rightarrow 0} y_{\text{outer}} = \lim_{x \rightarrow 0} e^{-x} = 1$$

and

$$\lim_{X \rightarrow \infty} y_{\text{inner}} = \lim_{X \rightarrow \infty} A_0(1 - e^{-X}) = A_0$$

thus, $A_0 = 1$. We have now determined our inner expansion at least to $O(1)$. We can now write down an expansion for $y(x; \varepsilon)$ at $O(1)$ that is valid for all $0 < x < 1$. This is

$$y_{\text{comp}} = y_{\text{inner}} + y_{\text{outer}} - \text{'common part'}$$

Our 'common part' in this example is just 1. Thus,

$$y_{\text{comp}} = e^{-x} + (1 - e^{-X}) - 1 = e^{-x} - e^{-x/\varepsilon}$$

Definition 2.7. (Van Dyke's method for matching) y_{outer} is a good approximation for $x = O(1)$ and y_{inner} is a good approximation for $x = O(\varepsilon)$.

Van Dyke: Take $x = O(\varepsilon)$ in y_{outer} (equivalently, write y_{outer} in terms of X and take $X = O(1)$). And write y_{inner} in

terms of $x(=\varepsilon X)$ with $x = O(1)$ then equate the two resulting expansions

$$\begin{aligned} y_{\text{outer}} &= e^{-x} + \varepsilon(1-x)e^{-x} + O(\varepsilon^2) \\ &= (1-x+x^2+\dots) + \varepsilon(1-x)(1-x+x^2+\dots) + O(\varepsilon^2) \end{aligned}$$

Now take $x = O(\varepsilon)$ then

$$\begin{aligned} &= 1-x+\varepsilon+O(\varepsilon^2) \\ y_{\text{inner}} &= A_0(1-e^{-X}) + \varepsilon(-A_0X - A_0Xe^{-X} + A_1(1-e^{-X})) + O(\varepsilon^2) \\ &= A_0(1-e^{-x/\varepsilon}) + \varepsilon(-A_0\frac{x}{\varepsilon} - A_0\frac{x}{\varepsilon}e^{-x/\varepsilon} + A_1(1-e^{-x/\varepsilon})) + O(\varepsilon^2) \end{aligned}$$

Note that as $\varepsilon \rightarrow 0$ with $x = O(1)$, $e^{-x/\varepsilon}$ is transcendentally small, i.e., $e^{-x/\varepsilon} = o(\varepsilon^p)$ for all $p > 0$. Then

$$= A_0 - A_0x + \varepsilon A_1 + O(\varepsilon^2)$$

Then equate the outer expansion and the inner expansion we have

$$A_0 = A_1 = 1$$

Now take the composite solution to be

$$\begin{aligned} y(x; \varepsilon) &= y_{\text{outer}} + y_{\text{inner}} - \text{"common part"} \\ &= e^{-x} + \varepsilon(1-x)e^{-x} + (1-e^{-x/\varepsilon}) + \varepsilon(-\frac{x}{\varepsilon} - \frac{x}{\varepsilon}e^{-x/\varepsilon} + (1-e^{-x/\varepsilon})) - (1-x+\varepsilon) + O(\varepsilon^2) \\ &= e^{-x} - (1+x)e^{-x/\varepsilon} + \varepsilon((1-x)e^{-x} - e^{-x/\varepsilon}) + O(\varepsilon^2) \end{aligned}$$

Note that Van Dyke's matching method does not always work.

Definition 2.8. (The intermediate variable method) Find (if it exists) a scaling function $\eta(\varepsilon)$ with $0 \ll \eta(\varepsilon) \ll 1$ as $\varepsilon \rightarrow 0$ such that y_{inner} and y_{outer} are both valid for $x = \eta(\varepsilon)\chi$ where $\chi = O(1)$. (i.e., $x = O(\eta(\varepsilon))$). Then write y_{inner} and y_{outer} in terms of $\eta(\varepsilon)\chi$ and equate them, and then equate coefficients of powers of ε and also equate coefficients of powers of $\eta(\varepsilon)$.

Example. $\eta(\varepsilon) = \varepsilon^p$ for $0 < p < 1$ works (we'll provide no proof of this here).

Then

$$\begin{aligned} y_{\text{outer}} &= e^{-x} + \varepsilon(1-e^{-x}) + O(\varepsilon^2) \\ &= e^{-\varepsilon^p\chi} + \varepsilon(1-x)e^{-\varepsilon^p\chi} + \dots \\ &= (1-\varepsilon^p\chi + O(\varepsilon^{2p})) + \varepsilon(1-\varepsilon^p\chi)(1+O(\varepsilon^p)) + O(\varepsilon^2) \end{aligned}$$

$$\begin{aligned} y_{\text{inner}} &= A_0(1-e^{-X}) + \varepsilon(-A_0X - A_0Xe^{-X} + A_1(1-e^{-X})) + \dots \\ &= A_0(1-e^{-\varepsilon^{p-1}\chi}) + \varepsilon(-A_0\varepsilon^{p-1}\chi - A_0\varepsilon^{p-1}\chi e^{-\varepsilon^{p-1}\chi} + A_1(1-e^{-\varepsilon^{p-1}\chi})) + \dots \end{aligned}$$

In fact, let us take $\frac{1}{2} < p < 1$ then

$$y_{\text{inner}} = A_0 + \varepsilon(-A_0\varepsilon^{p-1}\chi + A_1) + O(\varepsilon^2)$$

Equating these expressions for y_{inner} and y_{outer} gives: At $O(1)$, $A_0 = 1$. At $O(\varepsilon^p)$, $-A_0 = -1$. At $O(\varepsilon)$, $A_1 = 1$. (as found using Van Dyke's method).

2.4 Boundary layer location and interior layers

For general non-linear boundary value problems, boundary layers can only be found by trial and errors, however, in some cases, we can identify them, as follows.

Example. Consider the boundary value problem

$$\varepsilon y'' + p(x)y' + q(x)y = 0 \quad \text{on } 0 < x < 1$$

with $y(0) = A$, $y(1) = B$.

Consider the cases:

$$\begin{aligned} \text{Case 1 : } &\begin{cases} 1(a) : p(x) > 0 & \text{for } 0 \leq x \leq 1 \\ 1(b) : p(x) < 0 & \text{for } 0 \leq x \leq 1 \end{cases} \\ \text{Case 2 : } &p(x_*) = 0, p'(x_*) > 0 \end{aligned}$$

for some $0 < x_* < 1$, and x_* is the only simple root of $p(x)$ in $(0, 1)$. Case 3 is the same as case 2 but $p'(x_*) < 0$. Now consider Case 1: suppose there is a boundary layer at $x = x_0$ for some $0 \leq x_0 \leq 1$. Introduce $X = \frac{x-x_0}{\delta}$ for some δ with $\delta \rightarrow 0$ and $\varepsilon \rightarrow 0$. Taylor expansion of $p(x)$ and $q(x)$ about x_0 then

$$\begin{aligned} p(x) &= p(x_0) + p'(x_0)\delta X + \frac{\delta^2 X^2}{2}p''(x_0) + O(\delta^3) \\ q(x) &= q(x_0) + q'(x_0)\delta X + \frac{\delta^2 X^2}{2}q''(x_0) + O(\delta^3) \end{aligned}$$

Then our ODE becomes with $Y(X) = y(x_0 + \delta X)$ (write $\frac{d}{dx} = \frac{1}{\delta} \frac{d}{dX}$). Also assume $p(x)$ and $q(x)$ have Taylor expansion about x_0 then

$$\begin{aligned} p(x_0 + \delta X) &= p(x_0) + p'(x_0)\delta X + \frac{\delta^2 X^2}{2} p''(x_0) + O(\delta^3) \\ q(x_0 + \delta X) &= q(x_0) + q'(x_0)\delta X + \frac{\delta^2 X^2}{2} q''(x_0) + O(\delta^3) \end{aligned}$$

Then, our ODE can be written as

$$\frac{\varepsilon}{\delta^2} Y'' + (p(x_0) + \delta X p'(x_0) + \dots) \frac{Y'}{\delta} + (q(x_0) + \delta X q'(x_0) + \dots) Y = 0$$

Now take $\delta = \varepsilon$ then

$$Y'' + (p(x_0) + \varepsilon X p'(x_0) + \dots) Y' + \varepsilon (q(x_0) + \varepsilon X q'(x_0) + \dots) Y = 0$$

Seek Y of the form $Y(X; \varepsilon) = Y_0(X) + \varepsilon Y_1(X) + \dots$ then this gives at leading order

$$\begin{aligned} Y_0'' + p(x_0) Y_0' &= 0 \\ \implies Y_0(X) &= C_1 e^{-p(x_0)X} + C_2 \end{aligned}$$

for some constants C_1 and C_2 . For the case $p(x_0) > 0$, if there exists $x \in [0, 1]$ with $x < x_0$, then $X \rightarrow -\infty$ for this x as $\varepsilon \rightarrow 0$ so $Y_0(X)$ becomes exponentially large. A function $y(x)$ with this sort of behaviour local to x_0 generally does not have a boundary layer at x_0 . Given the above, if $p(x) > 0$ for all $x \in [0, 1]$ we can only have a boundary layer at $x = 0$. Similarly, if $p(x) < 0$ for $x \in [0, 1]$, we can only have a boundary layer at $x = 1$.

Case 2: $p(x_*) = 0$ for some $x_* \in (0, 1)$ with $p'(x_*) > 0$. As above, we have $Y_0(X) = C_1 e^{-p(x_0)X} + C_2$ for any $x_0 \in [0, 1]$. If $0 \leq x_0 < x_*$ then $p(x_0) < 0$ but there exists x with $x_0 < x < x_*$ and $X \rightarrow +\infty$ at this x as $\varepsilon \rightarrow 0$ so $Y_0(X)$ becomes exponentially large. Generally we do not expect to have a boundary layer at such an x_0 . Similarly, we do not expect to have a boundary layer at X_0 with $x_* < x_0 \leq 1$. So the only remaining possibility is of a boundary layer at $x = x_*$. Let us now examine y in the neighborhood of x_* with $X = \frac{x - x_*}{\delta}$ we find previously

$$\frac{\varepsilon}{\delta^2} Y'' + \frac{1}{\delta} (\delta X p'(x_*) + \dots) Y' + (q(x_*) + \delta X q'(x_*) + \dots) Y = 0$$

Now choose $\delta = \varepsilon^{1/2}$. Then we have

$$Y'' + (X p'(x_*) + \dots) Y' + (q(x_*) + \dots) Y = 0$$

To leading order, this is

$$Y'' + X p'(x_*) Y + q(x_*) Y = 0$$

Now seek Y of the form $Y(x; \varepsilon) = \varepsilon^m Y_0(X) + \dots$ for some m . Subbing this into the ODE for Y , we get at leading order:

$$Y_0'' + p'(x_*) X Y_0 + q(x_*) Y_0 = 0$$

this is known as the interior boundary layer equation. Let us now seek an outer solution: Recall,

$$\varepsilon y'' + p(x) y' + q(x) y = 0$$

seek y of the form $y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)$. This gives at leading order

$$p(x) y_0' + q(x) y_0 = 0$$

Recall also we required $y(0) = A$ and $y(1) = B$

$$\implies y_0(0) = A \text{ or } y_0(1) = B$$

then we have

$$y_{0l}(x) = A \exp\left(-\int_0^x \frac{q(t)}{p(t)} dt\right)$$

satisfies $y_0(0) = A$.

$$y_{0r}(x) = B \exp\left(\int_x^1 \frac{q(t)}{p(t)} dt\right)$$

satisfies $y_0(1) = B$. Let us examine the behaviour of y_{0l} and $y_{0r}(x)$ local to x_* . y_{0l} and $y_{0r}(x)$ are solutions to the equation

$$p(x) y_0' + q(x) y_0 = 0$$

Now by Taylor expansion of $p(x)$, $q(x)$ about x_* . More specifically, let

$$X = \frac{x - x_*}{\delta} \quad \delta \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

Also introduce $\hat{Y}_0 = y_0(x)$ then the above equation for y_0 becomes

$$(\delta X p'(x_*) + O(\delta^2)) \frac{\hat{Y}_0'}{\delta} + (q(x_*) + O(\delta)) \hat{Y}_0 = 0$$

At $O(1)$,

$$\begin{aligned} X p'(x_*) \hat{Y}_0' + q(x_*) \hat{Y}_0 &= 0 \\ \implies X \hat{Y}_0' + 2\kappa \hat{Y}_0 &= 0 \end{aligned}$$

where $\kappa = \frac{q(x_*)}{2p'(x_*)}$ then

$$\hat{Y}_0(X) = \exp\left(\int^x -\frac{2\kappa}{t} dt\right) = C |X|^{-2\kappa}$$

for some constant C .

$$\implies y_{ol}(x) = C_l |x - x_*|^{-2\kappa}, \quad y_{or}(x) = C_r |x - x_*|^{-2\kappa}$$

as $x \rightarrow x_*$. We want our inner solution to behave in such a way that it matches with these outer solutions. Recall now the Interior Boundary layer equation:

$$Y_0'' + p'(x_*)XY_0' + q(x_*)Y_0 = 0$$

where $X = \frac{x-x_*}{\varepsilon^{1/2}}$ and $Y(x) = \varepsilon^m Y_0(X) + \dots$. Define $\alpha = p'(x_*)$ and $\kappa = \frac{q(x_*)}{2p'(x_*)}$ then

$$Y_0'' + \alpha XY_0' + 2\kappa Y_0 = 0$$

Now introduce $z = -\frac{1}{2}\alpha X^2$ then the above equation becomes

$$z \frac{d^2 Y_0}{dz^2} + \left(\frac{1}{2} - z\right) \frac{dY_0}{dz} - \kappa Y_0 = 0$$

this is of the form of Kummer's equation:

$$z \frac{d^2 w}{dz^2} + (b - z) \frac{dw}{dz} - aw = 0$$

with $b = \frac{1}{2}$ and $a = \kappa$. The general solution of Kummer's equation:

$$w(z) = C_1 M(a, b, z) + C_2 z^{1-b} M(1 + a - b, z - b, z)$$

where $M(a, b, z)$ is the Kummer function, and C_1, C_2 are two arbitrary constants.

$$\implies Y_0(X) = A_0 M\left(\kappa, \frac{1}{2}, -\frac{1}{2}\alpha X^2\right) + B_0 X\left(\kappa + \frac{1}{2}, \frac{3}{2}, -\frac{1}{2}\alpha X^2\right)$$

for constant A_0 and B_0 . Let's now try to match this to our outer solutions. Consider Y_0 as $X \rightarrow \pm\infty$. Note, $\alpha = p'(x_*) > 0$ so $-\frac{1}{2}\alpha X^2 \rightarrow -\infty$ as $X \rightarrow \pm\infty$. Using known results of $M(a, b, z)$ we have

$$M\left(\kappa, \frac{1}{2}, -\frac{1}{2}\alpha X^2\right) \sim \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2} - \kappa)} \left(\frac{1}{2}\alpha X^2\right)^{-\kappa}$$

and

$$M\left(\kappa + \frac{1}{2}, \frac{3}{2}, -\frac{1}{2}\alpha X^2\right) \sim \frac{\Gamma(\frac{3}{2})}{\Gamma(1 - \kappa)} \left(\frac{1}{2}\alpha X^2\right)^{-\kappa-1/2}$$

Note that $(X^2)^{-\kappa} = |X|^{-2\kappa}$ and $(X^2)^{-\kappa-1/2} = |X|^{-2\kappa-1}$. We are taking branch of $(X^2)^{-\kappa}$ that is real and positive. Similarly for the branch of $(X^2)^{-\kappa-1/2}$. Then, as $X \rightarrow \pm\infty$ we have

$$\begin{aligned} Y_0(X) &= Y_0(X) = A_0 M\left(\kappa, \frac{1}{2}, -\frac{1}{2}\alpha X^2\right) + B_0 X\left(\kappa + \frac{1}{2}, \frac{3}{2}, -\frac{1}{2}\alpha X^2\right) \\ &\sim \frac{A_0 \sqrt{\pi}}{\Gamma(\frac{1}{2} - \kappa)} \left(\frac{1}{2}\alpha\right)^{-\kappa} |X|^{-2\kappa} \pm \frac{B_0 \sqrt{\pi} X}{2\Gamma(1 - \kappa)} \left(\frac{1}{2}\alpha\right)^{-\kappa-1/2} |X|^{-2\kappa} \\ &= \left(\frac{A_0}{\Gamma(\frac{1}{2} - \kappa)} \pm \frac{B_0}{\sqrt{2\alpha}\Gamma(1 - \kappa)}\right) \sqrt{\pi} \left(\frac{\alpha}{2}\right)^{-\kappa} |X|^{-2\kappa} \end{aligned}$$

Example. Find the leading order composite solution for $0 < \varepsilon \ll 1$ to the boundary value problem

$$\varepsilon y'' + (3x - 1)y' + xy = 0, \quad 0 < x < 1$$

with $y(0) = 1$ and $y(1) = 2$.

In this case, $p(x) = 3x - 1$ and $q(x) = x$. This boundary value problem is thus of the type Case 2, considered above, with $x_* = \frac{1}{3}$, $\alpha = p'(x_*) = 3$ and $\kappa = \frac{1}{18}$. The leading order equation for outer solution is

$$\begin{aligned} (3x - 1)y_0' + xy_0 &= 0 \\ \implies y_0(x) &= \frac{De^{-x/3}}{|3x - 1|^{1/9}} \end{aligned}$$

for constant D . D will be different depending on which boundary condition we impose. Fixing $y_0(0) = 1$ gives

$$y_{ol} = \frac{e^{-x/3}}{(1 - 3x)^{1/9}}$$

Fixing $y_0(1) = 2$ gives

$$y_{or} = \frac{2^{10/9} e^{(1-x)/3}}{(3x - 1)^{1/9}}$$

For the inner solution, with $\alpha = 3$ and $\kappa = \frac{1}{18}$, our earlier formula for $Y_0(X)$ gives

$$\begin{aligned} Y_0(X) &= A_0 M\left(\frac{1}{18}, \frac{1}{2}, -\frac{3}{2}X^2\right) + B_0 X M\left(\frac{5}{9}, \frac{3}{2}, -\frac{3}{2}X^2\right) \\ &\sim \left(\frac{A_0}{\Gamma(4/9)}\right) \pm \frac{B_0}{\sqrt{6}\Gamma(17/18)} \sqrt{\pi} \left(\frac{3}{2}\right)^{-1/18} |X|^{-1/9} \end{aligned}$$

as $X \rightarrow \pm\infty$. Now match the outer and inner solutions. Let us take the following limit of our outer solutions. Let $x = \frac{1}{3} + \varepsilon^{1/2}X$ where $X = O(1)$. Then

$$y_{ol}(x) = \frac{e^{-x/3}}{(1 - 3x)^{1/9}} = \frac{e^{-1/9} \cdot e^{-\varepsilon^{1/2}X/3}}{(-3\varepsilon^{1/2}X)^{1/9}}$$

$$= \frac{e^{-1/9}(1 + O(\varepsilon^{1/2}))}{3^{1/9}\varepsilon^{1/18}|X|^{1/9}} = \frac{e^{-1/9}}{3^{1/9}\varepsilon^{1/18}|X|^{1/9}}$$

Similarly,

$$y_{or}(x) = \frac{2^{10/9}e^{2/9}}{3^{1/9}\varepsilon^{1/18}|X|^{1/9}}$$

to leading order. Recall, our inner solution is of the form

$$Y(X) = \varepsilon^m Y_0(X) + \dots$$

in order for this to match with our outer solutions, we see from the above that we must have $m = -\frac{1}{18}$. Also, matching this inner solution with the outer solutions we get the following equations

$$\begin{aligned} \frac{e^{-1/9}}{3^{1/9}|X|^{1/9}} &= \left(\frac{A_0}{\Gamma(4/9)} - \frac{B_0}{\sqrt{6}\Gamma(17/18)} \right) \sqrt{\pi} \left(\frac{3}{2} \right)^{-1/18} |X|^{-1/9} \\ \frac{2^{10/9}e^{2/9}}{3^{1/9}|X|^{1/9}} &= \left(\frac{A_0}{\Gamma(4/9)} + \frac{B_0}{\sqrt{6}\Gamma(17/18)} \right) \sqrt{\pi} \left(\frac{3}{2} \right)^{-1/18} |X|^{-1/9} \end{aligned}$$

then we are able to find A_0 and B_0 . So our composite solution is

$$y(x) = \varepsilon^{-1/18} Y_0(X) + \begin{cases} y_{ol}(x) - y_{sl}(x) & x < 1/3 \\ y_{or}(x) - y_{sr}(x) & x > 1/3 \end{cases}$$

where

$$y_{sl}(x) = \frac{e^{-1/9}}{3^{1/9}\varepsilon^{1/18}|X|^{1/9}}$$

and

$$y_{sr}(x) = \frac{2^{10/9}e^{2/9}}{3^{1/9}\varepsilon^{1/18}|X|^{1/9}}$$

where $X = \frac{x - \frac{1}{3}}{\varepsilon^{1/2}}$.

Example. Consider the boundary value problem

$$\varepsilon y'' + xy' + 3x^3 y = 0 \quad x \in [-1, 1]$$

where $y(-1; \varepsilon) = -e$, $y(1; \varepsilon) = e^{-1}$ and $0 < \varepsilon \ll 1$.

There is a boundary layer at $x = 0$. First, seek outer solutions for $x < 0$ and $x > 0$ of the form $y = y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)$. Substitute this into ODE then at $O(1)$ we have

$$xy'_0 + 3x^3 y_0 = 0 \implies y_0(x) = Ae^{-x^3}$$

for some constant A .

$$\begin{aligned} y(-1; \varepsilon) = -e &\implies A = -1 \implies y_{ol}(x) = -e^{-x^3} \\ y(1; \varepsilon) = e^{-1} &\implies A = 1 \implies y_{or}(x) = e^{-x^3} \end{aligned}$$

Recall that

$$\begin{aligned} y_{ol} &\sim C_l |x - x_*|^{-2\kappa} \text{ as } x \uparrow x_* \\ y_{or} &\sim C_r |x - x_*|^{-2\kappa} \text{ as } x \downarrow x_* \end{aligned}$$

where $\kappa = \frac{q(x_*)}{2p'(x_*)}$ and C_l and C_r are constants. In this example, $q(x) = x^3$ and $x_* = 0$ so $q(x_*) = 0$ and hence $\kappa = 0$. We shall find the inner solution, let $X = \frac{x}{\delta(\varepsilon)}$ for some $\delta(\varepsilon) \ll 1$ and introduce $Y(X) = y(x) = y(\delta X)$. Then our ODE becomes

$$\begin{aligned} \frac{\varepsilon Y''}{\delta^2} + \frac{\delta XY'}{\delta} + 3(\delta X)^3 Y &= 0 \\ \implies \varepsilon Y'' + \delta^2 XY' + 3\delta^5 X^3 Y &= 0 \end{aligned}$$

For leading order balance, take $\delta = \varepsilon^{1/2}$ then we are left with (at leading order):

$$Y'' + XY' = 0$$

Seek $Y(X) = Y_0(X) + O(\varepsilon)$ then at leading order

$$\begin{aligned} Y_0'' + XY_0' &= 0 \\ \implies \frac{Y_0''}{Y_0'} &= -X \implies Y_0' = Be^{-x^2/2} \\ \implies Y_0 &= C + B \int_0^X e^{-s^2/2} ds \end{aligned}$$

for some constants B and C . We require

$$\begin{aligned} \lim_{x \uparrow 0} y_{ol}(x) &= \lim_{X \rightarrow -\infty} Y_0(X) \\ \implies -1 &= C + B \int_0^{-\infty} e^{-s^2/2} ds \\ \implies -1 &= C - B \sqrt{\frac{\pi}{2}} \end{aligned}$$

Similarly we must have

$$\lim_{x \downarrow 0} y_{or}(x) = \lim_{X \rightarrow +\infty} Y_0(X)$$

$$\begin{aligned} \implies 1 &= C + B\sqrt{\frac{\pi}{2}} \\ \implies C &= 0, \quad B = \sqrt{\frac{2}{\pi}} \end{aligned}$$

Thus,

$$Y_0(X) = \sqrt{\frac{2}{\pi}} \int_0^X e^{-s^2/2} ds$$

Finally, our leading order composite solution is for

$$\begin{aligned} y(x; \varepsilon) &= \begin{cases} y_{ol}(x) + Y_0(x) - \text{'common part'}, & x < 0 \\ y_{or}(x) + Y_0(x) - \text{'common part'}, & x > 0 \end{cases} \\ y(x; \varepsilon) &= \begin{cases} -e^{-x^3} + \sqrt{\frac{2}{\pi}} \int_0^{x/\varepsilon^{1/2}} e^{-s^2/2} ds + 1, & x < 0 \\ e^{-x^3} + \sqrt{\frac{2}{\pi}} \int_0^{x/\varepsilon^{1/2}} e^{-s^2/2} ds - 1, & x > 0 \end{cases} \end{aligned}$$

note that at $x = 0$ this solution is continuous and equal to 0.

2.5 Corner layers

Let us now consider case 2 but with $\kappa < 0$.

Example. Consider

$$\varepsilon y'' + (x^2 - \frac{1}{4})y - (x + \frac{1}{2})y = 0, \quad 0 < x < 1$$

where $y(0) = 2$, $y(1) = 3$.

Here $p(x) = x^2 - \frac{1}{4}$ which has a single simple zero in $(0, 1)$ at $x = \frac{1}{2}$ with $p'(\frac{1}{2}) = 1 = \alpha > 0$. Also, $q(x) = -(x + \frac{1}{2})$. Hence $\kappa = -\frac{1}{2}$. For the outer solution, seek $y(x, \varepsilon) = y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)$. At leading order

$$\begin{aligned} (x^2 - \frac{1}{4})y'_0 - (x + \frac{1}{2})y_0 &= 0 \\ \implies (x - \frac{1}{2})y'_0 - y_0 &= 0 \\ \implies y_0(x) &= A(x - \frac{1}{2}) \end{aligned}$$

for some constant A . $y(0) = 2 \implies A = -4$ so $y_{ol}(x) = -4x + 2$ and $y(1) = 3 \implies A = 6$ so $y_{or}(x) = 6x - 3$. For the inner solution: recall we introduce $X = \frac{x-x_*}{\varepsilon^{1/2}}$ and $Y(x) = y(x_* + \varepsilon^{1/2}X)$. Then we seek $Y(X) = \varepsilon^m Y_0(X) + \dots$ and $Y_0(X)$ satisfies IBLE:

$$Y_0'' + p'(x_*)XY_0 + q(x_*)Y_0 = 0$$

We get the solution:

$$Y_0(X) = A_0 M(\kappa, \frac{1}{2}, -\frac{1}{2}\alpha X^2) + B_0 X M(\kappa + \frac{1}{2}, \frac{3}{2}, -\frac{1}{2}\alpha X^2)$$

for some constants A_0 and B_0 with $\kappa = -\frac{1}{2}$ and $\alpha = 1$. Note that $M(0, b, z) \equiv 1$ then we have

$$Y_0(X) = A_0(-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}X^2) + B_0 X$$

We determine A_0 and B_0 by matching with $y_{ol}(x)$ and $y_{or}(x)$. First note,

$$Y_0(X) \sim (\frac{A_0}{\Gamma(1)} \pm \frac{B_0}{\sqrt{2}\Gamma(\frac{3}{2})})\sqrt{\pi}(\frac{1}{2})^{-1/2}|X|$$

as $X \rightarrow \pm\infty$. Now substitute $x = \frac{1}{2} + \varepsilon^{1/2}X$ into outer solution

$$y_{ol}(x) = 2 - 4x = 2 - 4(\frac{1}{2} + \varepsilon^{1/2}X) = -4\varepsilon^{1/2}X = 4\varepsilon^{1/2}|X|$$

$$y_{or}(x) = 6x - 3 = 6\varepsilon^{1/2}|X|$$

Recall now, $Y(X) = \varepsilon^m Y_0(X) + \dots$. Here, from the above form for y_{ol} and y_{or} in terms of X we see that $m = \frac{1}{2}$. Then we must have

$$\begin{aligned} 4 &= (A_0 - B_0\sqrt{\frac{2}{\pi}})\sqrt{2\pi} \\ 6 &= (A_0 + B_0\sqrt{\frac{2}{\pi}})\sqrt{2\pi} \end{aligned}$$

Here, our composite solution is

$$y(x) = A_0\varepsilon^{1/2}M(-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}X^2) + B_0\varepsilon^{1/2}X$$

where $X = \frac{x-\frac{1}{2}}{\varepsilon^{1/2}}$.

For case 3: $p(x)$ has a single simple zero in $(0, 1)$ at x_* but now $p'(x_*) < 0$. As for case 2, with $X = \frac{x-x_*}{\varepsilon^{1/2}}$ and $Y(X) = \varepsilon^m Y_0(X) + \dots$ we still get the IBLE for $Y_0(X)$ and its solution

$$Y_0(X) = A_0 M(\kappa, \frac{1}{2}, -\frac{1}{2}\alpha X^2) + B_0 X M(\kappa + \frac{1}{2}, \frac{3}{2}, -\frac{1}{2}\alpha X^2)$$

for constants A_0 and B_0 but not $\alpha = p'(x_*) < 0$. Then as $X \rightarrow \pm\infty$, so $-\frac{1}{2}\alpha X^2 \rightarrow +\infty$ so (see fact sheet for properties of $M(a, b, z)$) $Y_0(X)$ grows exponentially and thus is unsuitable for matching unless $A_0 = B_0 = 0$ and $Y_0(X) = 0$. However, we still need to satisfy the boundary conditions $y(0) = A$ and $y(1) = B$ where generally $A, B \neq 0$. Hence in fact, we will have boundary layers at $x = 0$ and 1 .

Example. Find the leading-order composite solution of the boundary value problem

$$\varepsilon y'' - (3x - 1)y' - xy = 0, \quad 0 < x < 1$$

with $y(0) = 1$ and $y(1) = 2$.

As for previous example, the general leading order outer solution is

$$y_0(x) = \frac{De^{-x/3}}{|3x - 1|^{1/9}}$$

for some constant D . However, in this case, unlike previous example we must take $D = 0$ as our solution about $x = \frac{1}{3}$ must be zero (the alternative is that it grows exponentially as $\varepsilon \rightarrow 0$). Then, in order to satisfy the boundary conditions at $x = 0$ and $x = 1$, we must have boundary layers at these points. Solution in boundary layer at $x = 0$ (compare with previous example). Let $X = \frac{x}{\varepsilon}$ and $Y(X) = y(\varepsilon X)$. Then

$$\varepsilon \frac{Y''}{\varepsilon^2} - (3\varepsilon X - 1) \frac{Y'}{\varepsilon} - \varepsilon XY = 0$$

i.e.,

$$Y'' - (3\varepsilon X - 1)Y' - \varepsilon^2 XY = 0$$

Now seek $Y(X)$ in the form

$$Y(X) = Y_0(X) + \varepsilon Y_1(X) + \dots$$

then $Y_0'' + Y_0' = 0$

$$\implies Y_0(X) = A_0 e^{-X} + B_0$$

for some constants A_0, B_0 . Use $y(0) = 1$ we have

$$Y_0(X) = A_0 e^{-X} + (1 - A_0)$$

Next, match this with the outer solution which is 0, we need

$$\lim_{X \rightarrow \infty} Y_0(X) = 0 \implies A_0 = 1$$

$$\implies Y_0(X) = e^{-X} = e^{-x/\varepsilon}$$

Solution in boundary layer at $x = 1$. Let $X = \frac{x-1}{\varepsilon}$ and also let $Y(X) = y(x)$ yields

$$\frac{\varepsilon Y''}{\varepsilon^2} - (2 + 3\varepsilon X) \frac{Y'}{\varepsilon} - (1 + \varepsilon X)Y = 0$$

$$Y'' - (2 + 3\varepsilon X)Y' - \varepsilon(1 + \varepsilon X)Y = 0$$

Seek $Y(X) = Y_0(X) + \varepsilon Y_1(X) + \dots$ then

$$\begin{aligned} Y_0'' - 2Y_0' &= 0 \\ Y_0(X) &= C_0 e^{2X} + D_0 \end{aligned}$$

for some constants C_0, D_0 . $y(1) = 2 \implies Y(0) = 2$ so

$$Y_0(X) = C_0 e^{2X} + (2 - C_0)$$

Now we match with the outer solution

$$\lim_{X \rightarrow -\infty} Y_0(X) = 0 \implies C_0 = 2$$

$$\implies Y_0(X) = 2e^{2X} = 2e^{2(x-1)/\varepsilon}$$

which means the composite solution is

$$y_{\text{comp}} = e^{-x/\varepsilon} + 2e^{2(x-1)/\varepsilon}$$

Example. (Triple deck boundary layers) Consider the following boundary value problems:

$$\varepsilon^3 y'' + x^3 y' - \varepsilon y = x^3, \quad 0 < x < 1$$

with $y(0) = 1$ and $y(1) = 3$. You are given that there is a boundary layer at $x = 0$. Find the leading order composite solution for $y(x)$.

For the outer solution, we seek $y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)$.

$$\implies \varepsilon^3(y_0'' + \varepsilon y_1' + \dots) + x^3(y_0' + \varepsilon y_1' + \dots) - \varepsilon(\dots) = x^3$$

$$y_0' = 1$$

which means $y(x) = x + A$ for some constant A . And, applying the condition $y(1) = 3$ gives $A = 2$ so

$$y_0(x) = x + 2$$

To find the inner layer, introduce $X = \frac{x}{\delta(\varepsilon)}$ for some $\delta(\varepsilon)$ such that $\delta(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$ and $Y(X) = y(x) = y(\delta X)$. Then

$$\varepsilon^3 \frac{Y''}{\delta^2} + (\delta X)^3 \frac{Y'}{\delta} - \varepsilon Y = (\delta X)^3$$

i.e.,

$$\frac{\varepsilon^3}{\delta^2} Y'' + \delta^2 X^3 Y' - \varepsilon Y = \delta^3 X^3$$

If $\frac{\varepsilon^3}{\delta^2} = \delta^2$ then $\delta = \varepsilon^{3/4}$. But then the order of these terms are $\varepsilon^{3/2}$ which is $\ll \varepsilon$ so in this case, we do not have leading order balance. Now, suppose $\delta = \varepsilon^{1/2}$, then $\frac{\varepsilon^3}{\delta^2} \ll \varepsilon$ so we have leading order balance in this case. Suppose $\frac{\varepsilon^3}{\delta^2} = \varepsilon$ then $\delta = \varepsilon$ and $\varepsilon^2 \ll \varepsilon$ so we have leading order balance in this case. To find the inner layer solution. For this, $\delta = \varepsilon^{1/2}$ then the equation becomes

$$\varepsilon Y'' + X^3 Y' - Y = \varepsilon^{1/2} X^3$$

Substitute $Y(X) = Y_0(X) + \varepsilon Y_1(X) + \dots$ yields

$$X^3 Y_0' - Y_0 = 0$$

$$\implies Y_0(X) = A e^{-1/2 X^2}$$

for some constant A . Inner-inner layer solution, for this $\delta = \varepsilon$ then the equation becomes

$$Y'' + \varepsilon X^3 Y' - Y = \varepsilon^2 X^3$$

Substitute $Y(X) = Y_0(X) + \varepsilon Y_1(X) + \dots$ yields

$$Y_0'' - Y_0 = 0$$

which gives solution $Y_0(X) = C e^{-X} + D e^X$ for some constants C and D . To match this with $Y_0(X)$ (our inner layer solution) we need to take $D = 0$ otherwise it would grow exponentially.

$$\implies Y_0(X) = C e^{-X}$$

Use $Y_0(0) = 1$ we have $C = 1$. To match the inner layer solution with the outer solution, we need

$$\lim_{X \rightarrow \infty} Y_0(X) = \lim_{t \rightarrow 0} y_0(x)$$

$$\implies A = 2$$

Therefore, our leading order composite solution is

$$y_{\text{comp}} = x + 2 + 2e^{-\varepsilon/2x^2} + e^{-x/\varepsilon} - 2$$

$$y_{\text{comp}} = x + 2e^{-\varepsilon/2x^2} + e^{-x/\varepsilon}$$

3 Examples from fluid dynamics

The governing equations for the velocity field $\underline{u}(\underline{x}, t)$ of an incompressible, Newtonian fluid are the Navier-Stokes equations, These are (in non-dimensional form).

$$\frac{\partial \underline{u}}{\partial t} + (\underline{u} \cdot \nabla) \underline{u} = -\nabla p + \varepsilon^2 \nabla^2 \underline{u}$$

and $\nabla \cdot \underline{u} = 0$ where $p = p(\underline{x}, t)$ is the associated pressure, and

$$\varepsilon^2 = \frac{1}{Re}$$

where Re is the Reynolds number of the flow. We'll consider Re large, i.e., $\varepsilon \ll 1$. This corresponds to flows for which the effects of viscosity are small. In $2D$, for a flow in the (x, y) -plane, with $\underline{u} = u(x, y, t)\underline{i} + v(x, y, t)\underline{j}$, we have

$$\begin{aligned} u_t + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{\partial p}{\partial x} + \varepsilon^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \\ v_t + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{\partial p}{\partial y} + \varepsilon^2 \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \end{aligned}$$

and

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

3.1 Flow past a flat plate

Consider the following flow past the plate. Assume the flow to be steady (so no time-dependence). $\underline{u} \rightarrow \underline{i}$ and $p \rightarrow 0$ as $x^2 + y^2 \rightarrow \infty$. Boundary conditions on the plate: $u = v = 0$ ("no-slip" conditions) on $y = 0$ and $0 \leq x \leq 1$. Symmetry implies $v = 0$ also for $y = 0$ and $x < 0, x > 1$ and $\frac{\partial u}{\partial y} = 0$ for $y = 0$ and $x < 0, x > 1$. We are going to derive an approximate solution for $\underline{u}$ for $\varepsilon \ll 1$.

(Outer expansion) Consider $x, y = O(1)$. Seek u, v and p of the form

$$\begin{aligned} u(x, y; \varepsilon) &= u_0(x, y) + O(\varepsilon) \\ v(x, y; \varepsilon) &= v_0(x, y) + O(\varepsilon) \\ p(x, y; \varepsilon) &= p_0(x, y) + O(\varepsilon) \end{aligned}$$

Substitute these expansions into Navier Stokes equations. At leading order, we get followings

$$\begin{aligned} u_0 \frac{\partial u_0}{\partial x} + v_0 \frac{\partial u_0}{\partial y} &= -\frac{\partial p_0}{\partial x} \\ u_0 \frac{\partial v_0}{\partial x} + v_0 \frac{\partial v_0}{\partial y} &= -\frac{\partial p_0}{\partial y} \\ \frac{\partial u_0}{\partial x} + \frac{\partial v_0}{\partial y} &= 0 \end{aligned}$$

Boundary conditions $u_0 \rightarrow 1, v_0 \rightarrow 0$ and $p_0 \rightarrow 0$ as $x^2 + y^2 \rightarrow \infty$. And $v_0 = 0$ for $y = 0$, $\frac{\partial u_0}{\partial y} = 0$ for $y = 0$ and $x < 0$ or $x > 1$. The above has the solution

$$u_0 = 1, \quad v_0 = 0, \quad p_0 = 0$$

(Inner expansion). Now consider $y = O(\delta)$ for some $\delta = \delta(\varepsilon) \ll 1$ but still with $x = O(1)$. Introduce $Y = \frac{y}{\delta}$ and write $u(x, y; \varepsilon) = U(x, Y) + O(\varepsilon)$ and $v(x, y; \varepsilon) = V_0(x, Y) + \varepsilon V(x, Y) + O(\varepsilon^2)$ and $p(x, y; \varepsilon) = P(x, Y) + O(\varepsilon)$. Then NS3 gives

$$\frac{\partial U}{\partial x} + O(\varepsilon) + \frac{1}{\delta} \frac{\partial}{\partial Y} (V_0 + \varepsilon V + O(\varepsilon^2)) = 0$$

At leading order this gives $\frac{\partial V_0}{\partial Y} = 0$

$$\implies V_0 = V_0(x)$$

But $v = 0$ for $y = 0$, hence $V_0(x)$ must equal 0 for all x . So there is no $O(1)$ term in our inner expansion for v . Now, NS1 gives

$$\begin{aligned} (U + O(\varepsilon)) \frac{\partial}{\partial x} (U + O(\varepsilon)) + (\varepsilon V + O(\varepsilon^2)) \frac{1}{\delta} \left(\frac{\partial}{\partial Y} (U + O(\varepsilon)) \right) &= -\frac{\partial}{\partial x} (P + O(\varepsilon)) + \varepsilon^2 \left(\frac{\partial^2}{\partial x^2} + \frac{1}{\delta^2} \frac{\partial^2}{\partial Y^2} \right) (U + O(\varepsilon)) \\ \implies \delta^2 U \frac{\partial U}{\partial x} + \dots + \delta \varepsilon \frac{\partial U}{\partial Y} + \dots &= -\delta^2 \frac{\partial P}{\partial x} + \varepsilon^2 \frac{\partial^2 U}{\partial Y^2} + \dots \end{aligned}$$

Now, pick $\delta = \varepsilon$. Then our leading term balance that contains the "viscous" term $\frac{\partial^2 U}{\partial Y^2}$

$$\implies U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial Y} = -\frac{\partial P}{\partial x} + \frac{\partial^2 U}{\partial Y^2}$$

And our NS2 gives

$$U \frac{\partial (\varepsilon V + \dots)}{\partial x} + \varepsilon V \frac{\partial (\varepsilon V + \dots)}{\partial Y} \frac{1}{\varepsilon} = -\frac{1}{\varepsilon} \frac{\partial P}{\partial Y} + \varepsilon^2 \left(\frac{\partial^2 (\varepsilon V + \dots)}{\partial x^2} + \frac{\partial^2 (\varepsilon V + \dots)}{\partial Y^2} \frac{1}{\varepsilon^2} \right)$$

At leading order, this gives

$$\frac{\partial P}{\partial Y} = 0 \implies P = P(x)$$

in the boundary layer. And our NS3 gives

$$\begin{aligned}\frac{\partial(U + \dots)}{\partial x} + \frac{1}{\varepsilon} \frac{\partial(\varepsilon V + \dots)}{\partial Y} &= 0 \\ \implies \frac{\partial U}{\partial x} + \frac{\partial V}{\partial Y} &= 0\end{aligned}$$

So the leading order equations in the boundary layer are:

$$\begin{aligned}U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial Y} &= -\frac{dP}{dx} + \frac{\partial^2 U}{\partial Y^2} \\ \frac{\partial U}{\partial x} + \frac{\partial V}{\partial Y} &= 0\end{aligned}$$

these are known as Prandtl's boundary layer equations. We will wish to match our inner solution with our outer solution. Prandtl's idea: the limit of the boundary layer solution as $Y \rightarrow \infty$ must be the same as the outer solution as $y \rightarrow 0$. This implies we must have $U \rightarrow 1$ as $Y \rightarrow \infty$. Note, the $O(1)$ outer and inner solutions for v are already matched: they are both 0). Finally, we must also have

$$\lim_{Y \rightarrow \infty} P(x) = 0$$

i.e. $P(x) = 0$. And the boundary conditions give:

$$U = V = 0$$

at $Y = 0, 0 \leq x \leq 1$. Then our symmetry condition also gives

$$V = \frac{\partial U}{\partial Y} = 0$$

at $Y = 0$ for $x < 0$ and $x > 1$. To solve the above for U and V , it is convenient to introduce the stream function $\Psi(x, Y)$ defined such that

$$U = \frac{\partial \Psi}{\partial Y}, \quad V = -\frac{\partial \Psi}{\partial x}$$

Then BL2 is satisfied automatically. Note BL1 is

$$U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial Y} = \frac{\partial^2 U}{\partial Y^2}$$

which becomes

$$\Psi_Y \Psi_{Yx} - \Psi_x \Psi_{YY} = \Psi_{YYY}$$

And $U \rightarrow 1$ as $Y \rightarrow \infty \implies \Psi_Y \rightarrow 1$ as $Y \rightarrow \infty$ while the conditions along $Y = 0$ become the following, along $Y = 0, V = 0 \implies \Psi_x = 0$ so $\Psi = 0$ along $Y = 0$. And $\Psi_Y = 0$ for $Y = 0, 0 \leq x \leq 1$ which is no-slip condition. $\Psi_{YY} = 0$ for $Y = 0$, for $x < 0$ and $x > 1$. We will now divide the boundary layer into 3 sections: $x < 0$ (upstream of the plate), $0 \leq x \leq 1$, and $x > 1$ (downstream of the plate). For $x < 0$, we have

$$\Psi_Y \Psi_{Yx} - \Psi_x \Psi_{YY} = \Psi_{YYY}$$

subject to $\Psi_Y \rightarrow 1$ as $Y \rightarrow \infty$ and $\Psi = \Psi_{yy} = 0$ along $Y = 0$. This has solution $\Psi = Y$ which gives $U = 1$ and $V = 0$. SO upstream of the plate, the leading order boundary layer solution is identical to the outer solution. For $0 \leq x \leq 1$, we have

$$\Psi_Y \Psi_{Yx} - \Psi_x \Psi_{YY} = \Psi_{YYY}$$

subject to $\Psi_Y \rightarrow 1$ as $Y \rightarrow \infty$ and $\Psi = \Psi_Y = 0$ along $Y = 0$ and, to coincide with the upstream solution, we must have $\Psi = Y$ at $x = 0$. We seek a solution for Ψ of the following ("self-similar") form: $\Psi(x, Y) = x^\alpha f(\eta)$ where $\eta = \frac{Y}{x^\beta}$ for some function f and constants α and β to be determined. Then we have

$$\Psi_Y = x^{\alpha-\beta} f'(\eta), \dots$$

Substituting inside the PDE gives

$$\begin{aligned}x^{2\alpha-2\beta-1}((\alpha-\beta)(f')^2 - \alpha f f'') &= x^{\alpha-3\beta} f'''(\eta) \\ \implies 2\alpha - 2\beta - 1 &= \alpha - 3\beta \\ \implies \alpha + \beta &= 1\end{aligned}$$

And then

$$(\alpha - \beta)(f')^2 - \alpha f' f'' = f'''$$

since $\Psi = \Psi_Y = 0 \implies f(0) = f'(0) = 0$ and $\Psi_Y \rightarrow 1$ as $Y \rightarrow \infty$

$$\implies x^{\alpha-\beta} f'(\infty) = 1 \implies \alpha - \beta = 0, f'(\infty) = 1$$

Thus, $\alpha = \beta = \frac{1}{2}$. Finally, note that the condition $\Psi = Y$ at $x = 0$ is equivalent to $x^\alpha f(\eta) = x^\beta \eta$ as $\eta \rightarrow \infty$. This is already satisfied since $\alpha = \beta = \frac{1}{2}$ and $f'(\infty) = 1$. To summarize, the solution in the boundary layer is of the form

$$\Psi(x, Y) = x^{1/2} f(\eta)$$

where $\eta = \frac{Y}{x^{1/2}}$ where f is determined by the boundary value problem

$$-\frac{1}{2} f' f'' = f'''$$

where $f(0) = f'(0) = 0$ and $f'(\infty) = 1$. This is known as the Blasius solution for the boundary layer. This boundary layer problem must be solved numerically.

4 Method of multiple scales

Example. Linear damped oscillator with weak damping:

$$y'' + 2\varepsilon y' + y = 0$$

subjected to the initial condition $y(0) = 1, y'(0) = 0$ and $0 < \varepsilon \ll 1$.

This has the following exact solution:

$$y_{\text{exact}} = e^{-\varepsilon t} [\cos(t\sqrt{1-\varepsilon^2}) + \frac{\varepsilon}{\sqrt{1-\varepsilon^2}} \sin(t\sqrt{1-\varepsilon^2})]$$

By Taylor expansion

$$\begin{aligned} \sqrt{1-\varepsilon^2} &= 1 + \frac{1}{2}(-\varepsilon^2) + \frac{1}{2}(-\frac{1}{2})(-\varepsilon^2)/\frac{1}{2!} + \dots = 1 - \frac{\varepsilon^2}{2} + \frac{\varepsilon^4}{8} + \dots \\ \Rightarrow y_{\text{exact}} &= e^{-\varepsilon t} [\cos(t - \frac{\varepsilon^2 t}{2} + O(\varepsilon^4 t)) + O(\varepsilon)] \end{aligned}$$

For $t = O(1)$, $e^{-\varepsilon t} = 1 + O(\varepsilon)$ so

$$y_{\text{exact}} = \cos t + O(\varepsilon)$$

So our $t = O(1)$, the effects of the damping are not yet evident (at leading order). But for $t = O(\varepsilon^{-1})$, we have $\varepsilon t = O(1)$ so

$$y_{\text{exact}} = e^{-\varepsilon t} \cos t + O(\varepsilon)$$

So we now detect amplitude modulation due to the damping. And on even longer time scales (e.g. $t = O(\varepsilon^{-2}), O(\varepsilon^{-4})$), we detect phase modulation (we have $\cos(t - \varepsilon^2 t/2)$) where we can no longer neglect $\varepsilon^2 t$ if $t = O(\varepsilon^{-2})$ etc. We would focus on the first two time scales $t = O(1)$ and $t = O(\varepsilon^{-1})$. We define a "fast" time variable $t_1 = t$ and "slow" time variable $t_2 = \varepsilon t$. Then on the time scale $t_2 = O(1)$ we have

$$y_{\text{exact}}(t) = e^{-t_2} \cos t_1 + O(\varepsilon)$$

Deriving asymptotic solution directly. First, let's seek an expansion of the "regular form":

$$y = Y_0(t) + \varepsilon Y_1(t) + O(\varepsilon^2)$$

Substitute this into ODE yields

$$(Y_0'' + \varepsilon Y_1'' + \dots) + 2\varepsilon(Y_0' + \varepsilon Y_1' + \dots) + (Y_0 + \varepsilon Y_1 + \dots) = 0$$

Then, at $O(1)$ we have $Y_0'' + Y_0 = 0$ which means

$$Y_0 = Ae^{it} + Be^{-it}$$

for some constants A and B . Use the initial conditions $Y_0(0) = 1$ and $Y_0'(0) = 0$ gives $A = B = \frac{1}{2}$ thus,

$$Y_0 = \frac{1}{2}(e^{it} + e^{-it}) = \cos t$$

Next, at $O(\varepsilon)$ we have

$$\begin{aligned} Y_1'' + 2Y_0' + Y_1 &= 0 \\ \Rightarrow Y_1'' + Y_1 &= -2Y_0' = -ie^{it} + ie^{-it} \\ Y_1(t) &= Ce^{it} + De^{-it} - Ate^{it} - Bte^{-it} \end{aligned}$$

So, we have

$$y = (Ae^{it} + Be^{-it}) + \varepsilon(Ce^{it} + De^{-it} - Ate^{it} - Bte^{-it}) + O(\varepsilon^2)$$

but if $t = O(\varepsilon^{-1})$ then the second term is in fact $O(1)$ so this expansion breaks down due to the presence of the terms $te^{\pm it}$ in $Y_1(t)$ which are no longer of $O(1)$ when $t = O(\varepsilon^{-1})$ (These terms called "secular terms"-they are ever-increasing in size). So this regular expansion is only correct for $t = O(1)$. For longer time scales, e.g. $t = O(\varepsilon)$ we need another expansion.

(Multiple-scale expansion) Define $t_1 = t$ and $t_2 = \varepsilon t$. Now seek y in the following form:

$$y = y_0(t_1, t_2) + \varepsilon y_1(t_1, t_2) + O(\varepsilon^2)$$

First, we rewrite our initial value problem in terms of t_1 and t_2 as follows. Note

$$\frac{d}{dt} = \frac{\partial t_1}{\partial t} \frac{\partial}{\partial t_1} + \frac{\partial t_2}{\partial t} \frac{\partial}{\partial t_2} = \frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2}$$

so

$$\frac{d^2}{dt^2} = (\frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2})(\frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2}) = \frac{\partial^2}{\partial t_1^2} + 2\varepsilon \frac{\partial^2}{\partial t_1 \partial t_2} + \varepsilon^2 \frac{\partial^2}{\partial t_2^2}$$

So our ODE becomes

$$(\frac{\partial^2 y}{\partial t_1^2} + 2\varepsilon \frac{\partial^2 y}{\partial t_1 \partial t_2} + \varepsilon^2 \frac{\partial^2 y}{\partial t_2^2}) + 2\varepsilon(\frac{\partial y}{\partial t_1} + \varepsilon \frac{\partial y}{\partial t_2}) + y = 0$$

substitute the expansion for y yields:

$$\frac{\partial^2 y_0}{\partial t_1^2} + \varepsilon \frac{\partial^2 y_1}{\partial t_1^2} + \dots + 2\varepsilon \frac{\partial^2 y_0}{\partial t_1 \partial t_2} + 2\varepsilon \frac{\partial y_0}{\partial t_1} + y_0 + \varepsilon y_1 + O(\varepsilon^2) = 0$$

At leading order $O(1)$ we have

$$\frac{\partial^2 y_0}{\partial t_1^2} + y_0 = 0$$

And at $O(\varepsilon)$, we have

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} - 2 \frac{\partial y_0}{\partial t_1}$$

Our initial conditions $y(0) = 1 \implies y_0(0, 0) + \varepsilon y_1(0, 0) + O(\varepsilon^2) = 1$. Then $y_0(0, 0) = 1$. Also $y'(0) = 0$ gives

$$\left(\frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2}\right)(y_0 + \varepsilon y_1 + O(\varepsilon^2))|_{t_1=t_2=0} = 0$$

At $O(1)$, this gives

$$\frac{\partial y_0}{\partial t_1}(0, 0) = 0$$

Solving the above ODE for y_0 subjected to these initial conditions gives

$$y_0(t_1, t_2) = A_0(t_2)e^{it_1} + A_1(t_2)e^{-it_1}$$

for some functions A_0 and A_1 . Our initial conditions give

$$A_0(0) + A_1(0) = 1$$

In addition,

$$\begin{aligned} iA_0(0) - iA_1(0) &= 0 \\ \implies A_0(0) &= A_1(0) = \frac{1}{2} \end{aligned}$$

Now consider the ODE we found at $O(\varepsilon)$,

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} - 2 \frac{\partial y_0}{\partial t_1}$$

Substitute the expression we have derived yields

$$\begin{aligned} \frac{\partial^2 y_1}{\partial t_1^2} + y_1 &= -2(iA_0' e^{it_1} - iA_1 e^{-it_1} + iA_0 e^{it_1} - iA_1 e^{-it_1}) \\ &= -2i(A_0' + A_0)e^{it_1} + 2i(A_1' + A_1)e^{-it_1} \end{aligned}$$

The solution for y_1 to this ODE contains a complementary function of the form $B_0(t_2)e^{it_1} + B_1(t_2)e^{-it_1}$ for some B_0 and B_1 , since $e^{\pm it_1}$ appear on the RHS of this equation, its particular integral will contain the terms $te^{\pm it_1}$ which will be ever increasing in magnitude as t increases unless we force A_0 and A_1 to be such that

$$A_0' + A_0 = 0, \quad A_1' + A_1 = 0$$

so that this RHS is 0. Solving these ODEs for A_0 and A_1 we get

$$A_0(t_2) = A_1(t_2) = \frac{1}{2}e^{-t_2}$$

Hence, at leading order, our two-scale expansion for $y(t; \varepsilon)$ is

$$\begin{aligned} y &= A_0(t_2)e^{it_1} + A_1(t_2)e^{-it_1} + O(\varepsilon) \\ &= \frac{e^{it_1} + e^{-it_1}}{2}e^{-t_2} + O(\varepsilon) \\ &= e^{-t_2} \cos t_1 + O(\varepsilon) = e^{-\varepsilon t} \cos t + O(\varepsilon) \end{aligned}$$

Example. (The van der Pol oscillator) Consider the initial value problem

$$y'' + \varepsilon y'(y^2 - 1) + y = 0$$

for $y(t; \varepsilon)$, with $0 < \varepsilon \ll 1$ and $y(0) = 5$, $y'(0) = 0$.

As in the previous example, if one seeks an expansion for y of the form

$$y(t; \varepsilon) = Y_0(t) + \varepsilon Y_1(t) + O(\varepsilon^2)$$

the solution for Y_1 contains the terms $te^{\pm it}$, and hence this expansion is only valid for $t = O(1)$; for $t = O(\varepsilon^{-1})$, it is no longer valid. This suggests seeking a multiple scale expansion for y , with 'slow' time variable εt . Let $t_1 = t$, $t_2 = \varepsilon t$, and seek an expansion for y of the form

$$y = y_0(t_1, t_2) + \varepsilon y_1(t_1, t_2) + O(\varepsilon^2)$$

As before, we have

$$\begin{aligned} \frac{d}{dt} &= \frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2} \\ \frac{d^2}{dt^2} &= \frac{\partial^2}{\partial t_1^2} + 2\varepsilon \frac{\partial^2}{\partial t_1 \partial t_2} + \varepsilon^2 \frac{\partial^2}{\partial t_2^2} \end{aligned}$$

Hence our ODE becomes

$$\begin{aligned} \frac{\partial^2}{\partial t_1^2} + 2\varepsilon \frac{\partial^2}{\partial t_1 \partial t_2} + \varepsilon^2 \frac{\partial^2}{\partial t_2^2} (y_0 + \varepsilon y_1 + O(\varepsilon^2)) &+ \varepsilon((y_0 + \varepsilon y_1 + O(\varepsilon^2)) - 1) \left(\frac{\partial}{\partial t_1} + \varepsilon \frac{\partial}{\partial t_2}\right) (y_0 + \varepsilon y_1 + O(\varepsilon^2)) \\ &+ (y_0 + \varepsilon y_1 + O(\varepsilon^2)) = 0 \end{aligned}$$

This gives at $O(1)$,

$$\frac{\partial^2 y_0}{\partial t_1^2} + y_0 = 0$$

At $O(\varepsilon)$ we have

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} - (y_0^2 - 1) \frac{\partial y_0}{\partial t_1}$$

The initial condition $y(0) = 5$ gives $y_0(0, 0) = 5$, and $y'(0) = 0$ gives

$$\left(\frac{\partial}{\partial t_2}\right)(y_0 + \varepsilon y_1 + O(\varepsilon^2))|_{t_1, t_2=0} = 0$$

At $O(1)$, this gives

$$\frac{\partial y_0}{\partial t_1}(0, 0) = 0$$

So our solution for y_0 is of the form

$$y_0(t_1, t_2) = A_0(t_2)e^{it_1} + \overline{A_0(t_2)}e^{-it_1}$$

for some $A_0(t_2)$ (using the fact that y is real and hence y_0 must be real valued). The above initial conditions on y_0 give

$$\begin{aligned} A_0(0) + \overline{A_0(0)} &= 5 \\ iA_0(0) - i\overline{A_0(0)} &= 0 \\ \implies A_0(0) &= \frac{5}{2} \end{aligned}$$

Now return to our ODE involving y_1 :

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} + (y_0^2 - 1) \frac{\partial y_0}{\partial t_1}$$

The solution to this will contain a complementary function of the form $B_0(t_2)e^{it_1} + \overline{B_0(t_2)}e^{-it_1}$ for some B_0 . The particular integral will contain the terms $te^{\pm it_1}$ which will make our expansion invalid for $t = O(\varepsilon^{-1})$ unless the term multiplying e^{it_1} on the RHS of this ODE to be zero. So, we are only interested in terms containing e^{it_1} on the RHS here, and not e^{int_1} for $n \neq 1$.

$$\begin{aligned} \frac{\partial^2 y_1}{\partial t_1^2} + y_1 &= -(2(iA_0' e^{it_1} - i\bar{A}_0' e^{-it_1})) + (iA_0 e^{it_1} - i\bar{A}_0 e^{-it_1})(A_0^2 e^{2it_1} + 2|A_0|^2 + \bar{A}_0^2 e^{-2it_1} - 1) \\ &= -(2iA_0' + iA_0(2|A_0|^2 - 1) - iA_0|A_0|^2)e^{it_1} + \dots \end{aligned}$$

So we require

$$2iA_0' + iA_0(2|A_0|^2 - 1) - iA_0|A_0|^2 = 0$$

i.e. $2A_0' + A_0|A_0|^2 - A_0 = 0$ which is known as the Landau-amplitude equation in this example. So we solve this equation for $A_0(t_2)$ as follows. Seek $A_0(t_2)$ of the form: $A_0 = r(t_2)e^{i\phi(t_2)}$ for some r and ϕ . Substituting this into the above equation for A_0 gives

$$\begin{aligned} 2(r' + i\phi' r)e^{i\phi} &= (r - r^3)e^{i\phi} \\ \implies 2(r' + i\phi' r) &= r - r^3 \end{aligned}$$

since every thing is real so $\phi'(t_2) = 0$ so ϕ is a constant. And real part gives $2r' = r - r^3$, multiply both sides by r yields

$$2r'r = r^2 - r^4$$

set $q = r^2$, $q' = q - q^2$ then

$$\begin{aligned} \log\left(\frac{q}{1-q}\right) &= t_2 + \tilde{C} \\ \implies q &= \frac{1}{1 - Ce^{-t_2}} \implies r = \frac{1}{\sqrt{1 - Ce^{-t_2}}} \end{aligned}$$

Thus,

$$A_0(t_2) = \frac{e^{i\phi}}{\sqrt{1 - Ce^{-t_2}}}$$

for some constant C and ϕ . Then $A_0(0) = \frac{5}{2}$ gives $\phi = 0$ and $C = \frac{21}{25}$. Hence, our leading order two-scale expansions for y is

$$y = y_0(t_1, t_2) + O(\varepsilon) = \frac{e^{it_1}}{\sqrt{1 - Ce^{-t_2}}} + \frac{e^{-it_1}}{\sqrt{1 - Ce^{-t_2}}} + O(\varepsilon) = \frac{2 \cos t}{\sqrt{1 - Ce^{-\varepsilon t}}} + O(\varepsilon)$$

where $C = \frac{21}{25}$.

Example. (Oscillator with non-linear elasticity) Consider the following boundary value problem:

$$y'' + y + \varepsilon y^2 = 0$$

subjected to the initial condition $y(0) = a$, $y'(0) = 0$ for some $y = y(t; \varepsilon)$ and $0 < \varepsilon \ll 1$.

Let us first seek a regular expansion for y , i.e.

$$y = y_0(t) + \varepsilon y_1(t) + \varepsilon^2 y_2(t) + O(\varepsilon^3)$$

Then

$$y_0'' + \varepsilon y_1'' + \varepsilon^2 y_2'' + \dots + y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \dots + \varepsilon y_0^2 + 2\varepsilon^2 y_0 y_1 + O(\varepsilon^3) = 0$$

At $O(1)$,

$$\begin{aligned} y_0'' + y_0 &= 0 \\ \implies y_0 &= Ae^{it} + \bar{A}e^{-it} \end{aligned}$$

for some constant A . Then at $O(\varepsilon)$,

$$\begin{aligned} y_1'' + y_1 &= -y_0^2 = -(A^2e^{2it} + \bar{A}^2e^{-2it} + 2|A|^2) \\ \implies y_1(t) &= Be^{it} + \bar{B}e^{-it} + Ce^{2it} + \bar{C}e^{-2it} + D \end{aligned}$$

for some constants B, C, D . Note, this does not contain any secular terms, so this regular expansion will still be valid for $t = O(\varepsilon^{-1})$. Next, at $O(\varepsilon^2)$,

$$y_2'' + y_2 = -2y_0y_1 = -2(Ae^{it} + \bar{A}e^{-it})(Be^{it} + \bar{B}e^{-it} + Ce^{2it} + \bar{C}e^{-2it} + D) = -2((AD + \bar{A}C)e^{it} + \dots)$$

one finds that this coefficient of e^{it} on the RHS here is non-zero. Hence, the solution for y_2 will contain secular terms and hence, the regular expansion will not be valid for $t = O(\varepsilon^{-2})$. Hence, seek a two-scale expansion for y in terms of $t_1 = t$ and $t_2 = \varepsilon^2 t$ of the form

$$y = y_0(t_1, t_2) + \varepsilon y_1(t_1, t_2) + \varepsilon^2 y_2(t_1, t_2) + O(\varepsilon^3)$$

Now,

$$\begin{aligned} \frac{d}{dt} &= \frac{\partial}{\partial t_1} + \varepsilon^2 \frac{\partial}{\partial t_2} \\ \frac{d^2}{dt^2} &= \frac{\partial^2}{\partial t_1^2} + 2\varepsilon^2 \frac{\partial^2}{\partial t_1 \partial t_2} + \varepsilon^4 \frac{\partial^2}{\partial t_2^2} \end{aligned}$$

so our ODE becomes

$$\left(\frac{\partial^2}{\partial t_1^2} + 2\varepsilon^2 \frac{\partial^2}{\partial t_1 \partial t_2} + \varepsilon^4 \frac{\partial^2}{\partial t_2^2}\right)(y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \dots) + (y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \dots) + \varepsilon(y_0^2 + 2\varepsilon y_0 y_1 + \dots) = 0$$

At $O(1)$,

$$\frac{\partial^2 y_0}{\partial t_1^2} + y_0 = 0 \implies y_0 = A_0(t_2)e^{it_1} + \overline{A_0(t_2)}e^{-it_1}$$

the initial condition $y(0) = a \implies y_0(0, 0) = a$ so $A_0(0) + \overline{A_0(0)} = a$. $y'(0) = 0 \implies \frac{\partial y_0}{\partial t_1}(0, 0) = 0$ so $iA_0(0) - i\overline{A_0(0)} = 0$ which means $A_0(0) = \overline{A_0(0)} = \frac{a}{2}$. At $O(\varepsilon)$ we have

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -y_0^2$$

where $y_0 = A_0e^{it} + \overline{A_0}e^{-it}$ so

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -(2|A_0|^2 + A_0^2e^{2it} + \overline{A_0}^2e^{-2it})$$

this has solution

$$y_1 = A_1(t_2)e^{it_1} + \overline{A_1(t_2)}e^{-it_1} - 2|A_0|^2 + \frac{1}{3}A_0^2e^{2it} + \frac{1}{3}\overline{A_0}^2e^{-2it}$$

note that this does not contain any secular terms. Then at $O(\varepsilon^2)$ we have

$$\frac{\partial^2 y_2}{\partial t_1^2} + y_2 = 2\frac{\partial^2 y_0}{\partial t_1 \partial t_2} - 2y_0y_1$$

We look for the coefficient of e^{it_1} of the RHS of the equation

$$-2\frac{\partial^2}{\partial t_1 \partial t_2}(A_0e^{it_1} + \overline{A_0}e^{-it_1}) - 2(A_0e^{it_1} + \overline{A_0}e^{-it_1})y_1 = -2iA_0'e^{it_1} + \dots - 2A_0e^{it_1}(-2|A_0|^2) - 2\overline{A_0}e^{-it_1}\frac{1}{3}A_0^2e^{2it_1}$$

so the coefficient of e^{it_1} is

$$-2iA_0' + 4A_0|A_0|^2 - \frac{2}{3}A_0|A_0|^2 = (-2iA_0' + \frac{10}{3}A_0|A_0|^2)$$

Hence, to prevent secular terms from arising in our solution for y_2 we set

$$iA_0' - \frac{5}{3}A_0|A_0|^2 = 0$$

which is the Landau equation and we solve this by seeking solutions of the form $A_0(t_2) = r(t_2)e^{i\phi(t_2)}$ for some real value functions r and ϕ . Then substituting this into this expression into our equation gives

$$\begin{aligned} i(r' + ri\phi')e^{i\phi} - \frac{5}{3}r^3e^{i\phi} &= 0 \\ \implies ir' - r\phi' - \frac{5}{3}r^3 &= 0 \end{aligned}$$

so $r' = 0 \implies r = \frac{a}{2}$ by our initial condition. Then when we look at the real parts

$$r\phi' + \frac{5}{3}r^3 = 0 \implies \phi' = -\frac{5}{12}a^2$$

in this case,

$$\phi = -\frac{5}{12}a^2t_2 + \phi_0$$

for some constant ϕ_0 . But since $A_0(0) = \frac{a}{2}$ so $\phi_0 = 0$. Hence,

$$A_0(t_2) = \frac{a}{2} \exp\left(-\frac{5ia^2t_2}{12}\right)$$

Hence, at leading order our two scale expansion for y is

$$\begin{aligned} y &= \frac{a}{2} \exp\left(-\frac{5ia^2t_2}{12}\right)(e^{it_1}) + \frac{a}{2} \exp\left(\frac{5ia^2t_2}{12}\right)(e^{-it_1}) + O(\varepsilon) \\ &= a \cos\left(t_1 - \frac{5a^2}{12}t_2\right) + O(\varepsilon) = \cos\left(t - \frac{5a^2}{12}\varepsilon^2t\right) + O(\varepsilon) \end{aligned}$$

So the oscillations have constant amplitude but over a time-scale of $t = O(\varepsilon^{-2})$ they also exhibit a change in phase.

Example. (Forced van der Pol oscillator) Consider the following boundary value problem for a real-valued function $y(t; \varepsilon)$:

$$y'' + \varepsilon(y^2 - 1)y' + y = Q \cos(\omega t)$$

with $y(0) = a$, $y'(0) = 0$ and Q, ω, a are real constants with $Q, a = O(1)$ and $\omega \notin \mathbb{Z}$. Define fast and slow time variables $t_1 = t$, $t_2 = \varepsilon t$ respectively. We seek a solution for y of the form

$$y = y_0(t_1, t_2) + \varepsilon y_1(t_1, t_2) + O(\varepsilon^2)$$

so our ODE becomes

$$\left(\frac{\partial^2}{\partial t_1^2} + 2\varepsilon \frac{\partial^2}{\partial t_1 \partial t_2} + \dots\right)(y_0 + \varepsilon y_1 + \dots) + \varepsilon(y_0^2 - 1) \frac{\partial y_0}{\partial t_1} + (y_0 + \varepsilon y_1) + O(\varepsilon^2) = Q \cos(\omega t_1)$$

At $O(1)$,

$$\frac{\partial^2 y_0}{\partial t_1^2} + y_0 = Q \cos(\omega t)$$

At $O(\varepsilon)$ we have

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} + (1 - y_0^2) \frac{\partial y_0}{\partial t_1}$$

And the initial conditions are

$$y(0) = a \implies y_0(0, 0) = a, \quad y'(0) = 0 \implies \frac{\partial y_0}{\partial t_1}(0, 0) = 0$$

Solution for y_0 of the homogeneous equation is given by

$$y_0 = A(t_2)e^{it_1} + \overline{A(t_2)}e^{-it_1}$$

for some $A(t_2)$. To find the particular integral we need $y_0 = B \cos(\omega t)$ so

$$(-\omega^2 + 1)B \cos(\omega t_1) = Q \cos(\omega t_1) \implies B = \frac{Q}{1 - \omega^2}$$

So,

$$y_0(t_1, t_2) = A(t_2)e^{it_1} + \overline{A(t_2)}e^{-it_1} + Ce^{i\omega t_1} + C\overline{e^{i\omega t_1}}$$

where $C = \frac{Q}{2(1-\omega^2)} \in \mathbb{R}$. And the boundary condition gives

$$\begin{aligned} A(0) + \overline{A} + 2C &= a, \quad iA(0) - i\overline{A} + i\omega C - i\omega C = 0 \\ \implies A(0) &= \frac{a}{2} - C = \frac{1}{2}\left(a - \frac{Q}{1 - \omega^2}\right) \end{aligned}$$

Now look at the equation at $O(\varepsilon)$

$$\frac{\partial^2 y_1}{\partial t_1^2} + y_1 = -2 \frac{\partial^2 y_0}{\partial t_1 \partial t_2} + (1 - y_0^2) \frac{\partial y_0}{\partial t_1}$$

We wish to eliminate terms proportional to e^{it_1} from the RHS of this equation. Given the above form for y_0 , this RHS is (terms containing e^{it_1})

$$\begin{aligned} &-2iA'e^{it_1} + iAe^{it_1} - iA(2|A|^2 + 2C^2)e^{it_1} + iA|A|^2e^{it_1} + 2iAC^2\omega e^{it_1} - 2iAC^2\omega e^{it_1} \\ &= -2iA'e^{it_1} + iAe^{it_1} - iA(2|A|^2 + 2C^2)e^{it_1} + iA|A|^2e^{it_1} = ie^{it_1}(-2A' + A - A|A|^2 - 2C^2A) \end{aligned}$$

We set this to be zero. Thus, our Landan equation is

$$2A' = -A|A|^2 - 2C^2A + A$$

Seek $A(t_2) = r(t_2)e^{i\phi(t_2)}$ for some real valued function $r(t_2)$, $\phi(t_2)$ then

$$\begin{aligned} 2(r' + i\phi'r)e^{i\phi} &= re^{i\phi}(1 - r^2 - 2C^2) \\ \implies 2(r' + i\phi'r) &= r(1 - r^2 - 2C^2) \end{aligned}$$

when we look at the imaginary part, $\phi' = 0 \implies \phi$ is a constant. Then $\phi = 0$ since $A(0) \in \mathbb{R}$. And the real part gives

$$2r' = r(1 - r^2 - 2C^2) \implies 2rr' = r^2(1 - r^2 - C^2)$$

let $q = r^2$ then

$$\begin{aligned} q' &= q(1 - 2C^2 - q) \implies \int dt_2 = \int \frac{dq}{q(1 - 2C^2 - q)} \\ \implies t_2 &= \frac{1}{1 - 2C^2} \int \left(\frac{1}{q} + \frac{1}{1 - 2C^2 - q}\right) dq = \frac{1}{1 - 2C^2} \log\left(\frac{q}{1 - 2C^2 - q}\right) + \tilde{D} \end{aligned}$$

for some constant $\tilde{D}$. Let $\beta = 1 - 2C^2$ then

$$\frac{\beta - q}{q} = De^{-\beta t_2}$$

for some constants D . Thus,

$$q = \frac{\beta}{1 + De^{-\beta t_2}}$$

and $q(0) = (A(0))^2$ so

$$D = \frac{\beta}{(A(0))^2} - 1$$

$$\Rightarrow A(t_2) = A(0) \sqrt{\frac{\beta}{A(0)^2 + (\beta - A(0)^2)e^{-\beta t_2}}}$$

5 The WKB method

Example. Consider the following initial value problem:

$$\varepsilon^2 y'' + (\tanh(x-3))y = 0$$

with $y(0) = 10^{-5}$, $y'(0) = 0$ for $y = y(x; \varepsilon)$ and $0 < \varepsilon \ll 1$.

Thus exhibits oscillatory behaviour for $x > 3$ and monotonic growth in the interval $0 < x < 3$. Note that $\tanh(x-3)$ changes sign at $x = 3$. The WKB method can be used to solve such problems where the solutions exhibit oscillatory behaviour in one region, and monotonic (in fact, exponential) growth in a neighbouring region. Such problems arise, for example, in wave propagation through a variable medium.

5.1 The canonical equation

Consider the following equation:

$$\varepsilon^2 y'' - q(x)y = 0$$

for $y = y(x; \varepsilon)$ and some function $q(x)$ where $0 < \varepsilon \ll 1$.

Example. In the case $q = \text{constant}$, this has solution:

$$y(x; \varepsilon) = \begin{cases} Ae^{\sqrt{q}x/\varepsilon} + Be^{-\sqrt{q}x/\varepsilon} & \text{for } q > 0 \\ Ax + B & \text{for } q = 0 \\ Ae^{i\sqrt{-q}x/\varepsilon} + Be^{-i\sqrt{-q}x/\varepsilon} & \text{for } q < 0 \end{cases}$$

for some constants A and B . So we have exponential growth/decay for $q > 0$, but oscillatory behaviour for $q < 0$ (and in this second case, the period of these oscillation is $2\pi\varepsilon/\sqrt{-q}$, which obviously $\rightarrow 0$ as $\varepsilon \rightarrow 0$). This suggests that in the general case, for $q(x)$ non-constant, the nature of the solution for y (either oscillatory or exponentially growing/decaying) depends on the sign of $q(x)$.

We are going to seek solutions to this canonical equation in the WKB form:

$$y(x) = (y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots) \exp\left(\frac{\theta(x)}{\varepsilon}\right)$$

where the function $\theta(x)$ (to be determined) is known as the phase function. This gives

$$y'(x) = \left(\frac{y_0\theta'}{\varepsilon} + \frac{\varepsilon y_1\theta'}{\varepsilon} + y_0' + O(\varepsilon)\right) \exp\left(\frac{\theta}{\varepsilon}\right)$$

and so

$$y'' = \left(\frac{y_0(\theta')^2}{\varepsilon^2} + \frac{y_1(\theta')^2}{\varepsilon} + \frac{y_0'\theta'}{\varepsilon} + \frac{y_0''}{\varepsilon} + O(1)\right) \exp\left(\frac{\theta}{\varepsilon}\right) = \left(\frac{y_0(\theta')^2}{\varepsilon^2} + \frac{y_0'' + 2y_0\theta' + y_1(\theta')^2}{\varepsilon} + O(1)\right) \exp\left(\frac{\theta}{\varepsilon}\right) =$$

Substituting the above expressions into our ODE gives, at $O(1)$

$$\begin{aligned} y_0(\theta')^2 - qy_0 &= 0 \\ \implies \theta' = \pm q^{1/2} &\implies \theta(x) = \pm \int^x (q(t))^{1/2} dt \end{aligned}$$

which is known as the Eikonal equation. At $O(\varepsilon)$ we have

$$y_0\theta'' + 2y_0\theta' + y_1(\theta')^2 - qy_1 = 0$$

Since $(\theta')^2 - q = 0$ so

$$y_0\theta'' + 2y_0'\theta' = 0$$

which is known as the transport equation.

$$\begin{aligned} \implies \frac{y_0'}{y_0} &= -\frac{\theta''}{2\theta'} \\ \implies y_0 &= B(\theta')^{-1/2} \end{aligned}$$

for some constant B which means $y_0 = B(q(x))^{-1/4}$. This indicates the leading order WKB solution to our ODE is

$$y_{\text{WKB}}(x) = \frac{1}{q(x)^{1/4}} \left(C_1 \exp\left(\frac{1}{\varepsilon} \int_{x_0}^x (q(t))^{1/2} dt\right) + C_2 \exp\left(-\frac{1}{\varepsilon} \int_{x_0}^x (q(t))^{1/2} dt\right) \right)$$

for some constants C_1 and C_2 and x_0 can be chosen freely without loss of generality. This form for y_{WKB} is valid in regions where $q < 0$ as well as regions where $q > 0$. However, for x such that $q < 0$, it is sometimes more convenient to write

$$y_{\text{WKB}}(x) = \frac{1}{|q(x)|^{1/4}} \left(\tilde{C}_1 \exp\left(\frac{i}{\varepsilon} \int_{x_0}^x (-q(t))^{1/2} dt\right) + \tilde{C}_2 \exp\left(-\frac{i}{\varepsilon} \int_{x_0}^x (-q(t))^{1/2} dt\right) \right)$$

for arbitrary constants $\tilde{C}_1$ and $\tilde{C}_2$.

5.2 Turning points

Let us examine what happens in regions where the solution changes from being exponential to oscillatory, i.e. regions local to points where $q(x) = 0$. These are called turning point. Let's suppose $q(x_*) = 0$ and $q'(x_*) > 0$. Let

$$X = \frac{x - x_*}{\gamma \varepsilon^p}$$

for some p to be determined and some constant γ . In addition, let $y(X) = y(x) = y(x_* + \gamma \varepsilon^p X)$. We are going to consider $X = O(1)$ i.e. x local to x_* . In this region we approximate $q(x)$ by

$$q(x) = q'(x_*)(x - x_*) = q'(x_*)\gamma \varepsilon^p X$$

Now, in terms of X and Y our ODE becomes

$$\frac{\varepsilon^2}{\gamma^2 \varepsilon^{2p}} Y'' - q'(x_*)\gamma \varepsilon^p X = 0$$

$$\implies Y'' - q'(x_*)\gamma^3 \varepsilon^{3p-2} XY = 0$$

Pick $p = \frac{2}{3}$ in order to have leading order balance. Also, pick $\gamma = (q'(x_*))^{-1/3}$ then we have

$$Y'' - XY = 0$$

which is known as the Airy's equation. The general solution of this is

$$Y(X) = \alpha Ai(X) + \beta Bi(X)$$

for arbitrary real constants α and β where $Ai(X)$ and $Bi(X)$ are the so-called Airy functions. Then

$$y_{\text{WKB}}(x) = \begin{cases} \frac{1}{|q(x)|^{1/4}} (\tilde{C}_1 \exp\left(\frac{i}{\varepsilon} \int_{x_*}^x |q(x)|^{1/2} dt\right) + \tilde{C}_2 \exp\left(-\frac{i}{\varepsilon} \int_{x_*}^x |q(x)|^{1/2} dt\right)) & x - x_* \ll -\varepsilon^{2/3} \\ \alpha Ai(X) + \beta Bi(X) \text{ where } X = \frac{x - x_*}{(q'(x_*)^{1/3}) \varepsilon^{2/3}} & x - x_* = O(\varepsilon^{2/3}) \\ \frac{1}{|q(x)|^{1/4}} (C_1 \exp\left(\frac{1}{\varepsilon} \int_{x_*}^x |q(x)|^{1/2} dt\right) + C_2 \exp\left(-\frac{1}{\varepsilon} \int_{x_*}^x |q(x)|^{1/2} dt\right)) & x - x_* \gg \varepsilon^{2/3} \end{cases}$$

We now need to match these solutions. Doing so we will determine some of C_1 , C_2 , $\tilde{C}_1$, $\tilde{C}_2$, α and β . We'll match with the 'left' outer solution by considering the the solution as $x \uparrow x_*$, and consider the inner solution as $X \rightarrow +\infty$ and find the 'right' outer solution as $x \downarrow x_*$. First consider the left solution as $x \uparrow x_*$. Approximate $q(x)$ by $q(x) = q'(x_*)(x - x_*)$. Then we have

$$y_{\text{WKB}}(x) = \frac{1}{(q'(x_*)(x_* - x))^{1/4}} x (\tilde{C}_1 \exp\left(\frac{i}{\varepsilon} \int_{x_*}^x (q'(x_*)(x_* - x))^{1/2} dt\right) + \tilde{C}_2 \exp\left(-\frac{i}{\varepsilon} \int_{x_*}^x (q'(x_*)(x_* - x))^{1/2} dt\right))$$

Note that

$$\int_{x_*}^x (x_* - t)^{1/2} dt = -\frac{2}{3} (x_* - x)^{3/2}$$

$$y_{\text{WKB}}(x) = \frac{1}{(q'(x_*)(x_* - x))^{1/4}} [\tilde{C}_1 \exp\left\{\frac{i(q'(x_*))^{1/2}}{\varepsilon} \cdot (-\frac{2}{3} (x_* - x)^{3/2})\right\} + \tilde{C}_2 \exp\left\{\frac{i(q'(x_*))^{1/2}}{\varepsilon} \cdot (\frac{2}{3} (x_* - x)^{3/2})\right\}]$$

Note that in terms of $X = \frac{x - x_*}{(q'(x_*)^{-1/3}) \varepsilon^{2/3}}$ as $X \rightarrow -\infty$ we have

$$y_{\text{WKB}} = \frac{1}{[q'(x_*)(q'(x_*))^{-1/3} \varepsilon^{2/3} |X|]^{1/4}} \times [\tilde{C}_1 \exp\left\{-\frac{2i|X|^{3/2}}{3}\right\} + \tilde{C}_2 \exp\left\{\frac{2i|X|^{3/2}}{3}\right\}]$$

$$= \frac{1}{(q'(x_*)\varepsilon)^{1/6} |X|^{1/4}} \times [\tilde{C}_1 \exp\left\{-\frac{2i|X|^{3/2}}{3}\right\} + \tilde{C}_2 \exp\left\{\frac{2i|X|^{3/2}}{3}\right\}], \quad X \rightarrow -\infty$$

Similarly, considering the right outer solution, one finds

$$y_{\text{WKB}} = \frac{1}{(q'(x_*)\varepsilon)^{1/6} X^{1/4}} [C_1 \exp\left\{\frac{2}{3} X^{3/2}\right\} + C_2 \exp\left\{-\frac{2}{3} X^{3/2}\right\}], \quad X \rightarrow +\infty$$

Now consider limits of the inner solution, we have

$$Y(X) = \alpha Ai(X) + \beta Bi(X) \sim \frac{\alpha}{\sqrt{\pi}|X|^{1/4}} \cos\left(\frac{2}{3}|X|^{3/2} - \frac{\pi}{4}\right) + \frac{\beta}{\sqrt{\pi}|X|^{1/4}} \cos\left(\frac{2}{3}|X|^{3/2} + \frac{\pi}{4}\right), \quad X \rightarrow -\infty$$

$$\sim \frac{\alpha}{2\sqrt{\pi}X^{1/4}} \exp\left\{-\frac{2}{3}X^{3/2}\right\} + \frac{\beta}{\sqrt{\pi}X^{1/4}} \exp\left\{\frac{2}{3}X^{3/2}\right\}, \quad X \rightarrow +\infty$$

So, by comparing the limits as $X \rightarrow +\infty$ we see that

$$\frac{\alpha}{2\sqrt{\pi}} = \frac{C_2}{(q'(x_*)\varepsilon)^{1/6}}, \quad \frac{\beta}{\sqrt{\pi}} = \frac{C_1}{(q'(x_*)\varepsilon)^{1/6}}$$

By comparing limits as $X \rightarrow -\infty$, one finds 2 further relations, now between α , β , $\tilde{C}_1$, $\tilde{C}_2$.

Example. Using the WKB method to solve the following boundary value problem:

$$\varepsilon^2 y'' - (2 \tanh x) y = 0$$

with $y(-1) = a$ for some constant a and $y \rightarrow 0$ as $x \rightarrow +\infty$ for $x \in [-1, +\infty)$ and $0 < \varepsilon \ll 1$.

This equation is of the above canonical type with $q(x) = 2 \tanh(x)$. Note, this has single turning point at $x = 0$ with

$q'(0) > 0$. For $x > 0$,

$$y_{\text{WKB}}(x) = \frac{1}{(q(x))^{1/4}} [C_1 \exp\left\{\frac{1}{\varepsilon} \int_0^x (q(t))^{1/2} dt\right\} + C_2 \exp\left\{-\frac{1}{\varepsilon} \int_0^x (q(t))^{1/2} dt\right\}]$$

so the boundary condition implies $C_1 = 0$. So our right outer solution reduces to

$$y_{\text{WKB}}(x) = \frac{C_2}{(q(x))^{1/4}} \exp\left\{-\frac{1}{\varepsilon} \int_0^x (q(t))^{1/2} dt\right\}$$

As $x \downarrow 0$, we can write

$$q(x) \equiv 2 \tanh x = 2x + O(x^3)$$

so we get

$$y_{\text{WKB}}(x) = \frac{C_2}{(2x)^{1/4}} \exp\left\{-\frac{1}{\varepsilon} \int_0^x (2t)^{1/2} dt\right\} = \frac{C_2}{(2x)^{1/4}} \exp\left\{-\frac{1}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} x^{3/2}\right\}$$

For $x < 0$, our left outer solution is of the form

$$y_{\text{WKB}}(x) = \frac{1}{|q(x)|^{1/4}} [\tilde{C}_1 \exp\left\{\frac{i}{\varepsilon} \int_0^x |q(t)|^{1/2} dt\right\} + \tilde{C}_2 \exp\left\{-\frac{i}{\varepsilon} \int_0^x |q(t)|^{1/2} dt\right\}]$$

as $x \uparrow 0$, this becomes

$$y_{\text{WKB}}(x) = \frac{1}{(-2x)^{1/4}} [\tilde{C}_1 \exp\left\{-\frac{i}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} (-x)^{3/2}\right\} + \tilde{C}_2 \exp\left\{\frac{i}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} (-x)^{3/2}\right\}]$$

Our inner solution is

$$Y(X) = \alpha A i(X) + \beta B i(X)$$

where $X = \frac{x}{(q'(0))^{-1/3} \varepsilon^{2/3}} = \frac{x}{2^{-1/3} \varepsilon^{2/3}}$ and $Y(X) = y(x) = y(2^{-1/3} \varepsilon^{2/3} X)$. As $X \rightarrow +\infty$, this becomes

$$Y \sim \frac{\alpha}{2\sqrt{\pi} X^{1/4}} \exp\left\{-\frac{2}{3} X^{3/2}\right\} + \frac{\beta}{\sqrt{\pi} X^{1/4}} \exp\left\{\frac{2}{3} X^{3/2}\right\}$$

Recall now, our limit as $x \downarrow 0$ of our outer solution is

$$\begin{aligned} y_{\text{WKB}}(x) &= \frac{C_2}{(2x)^{1/4}} \exp\left\{-\frac{1}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} x^{3/2}\right\} = \frac{C_2}{(2(2^{-1/3} \varepsilon^{2/3}) X)^{1/4}} \exp\left\{-\frac{2X^{3/2}}{3}\right\} \\ &= \frac{C_2}{(2\varepsilon)^{1/6} X^{1/4}} \exp\left\{-\frac{2}{3} X^{3/2}\right\} \end{aligned}$$

So, comparing this with the limit as $X \rightarrow +\infty$ of our inner solution, we see

$$\beta = 0, \quad \frac{\alpha}{2\sqrt{\pi}} = \frac{C_2}{(2\varepsilon)^{1/6}}$$

Now, as $X \rightarrow -\infty$ our inner solution becomes

$$Y = \frac{\alpha}{\sqrt{\pi} |X|^{1/4}} \cos\left(\frac{2}{3} |X|^{3/2} - \frac{\pi}{4}\right)$$

and as $x \uparrow 0$ our left outer solution is

$$\begin{aligned} y_{\text{WKB}}(x) &= \frac{1}{(-2x)^{1/4}} [\tilde{C}_1 \exp\left\{-\frac{i}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} (-x)^{3/2}\right\} + \tilde{C}_2 \exp\left\{\frac{i}{\varepsilon} \cdot \frac{2\sqrt{2}}{3} (-x)^{3/2}\right\}] \\ &= \frac{1}{(2\varepsilon)^{1/6} |X|^{1/4}} [\tilde{C}_1 \exp\left\{-\frac{2i}{3} |X|^{3/2}\right\} + \tilde{C}_2 \exp\left\{\frac{2i}{3} |X|^{3/2}\right\}] \end{aligned}$$

Comparing this with the limit as $X \rightarrow -\infty$ of our inner solution, one finds

$$\frac{\alpha e^{-i\pi/4}}{2\sqrt{\pi}} = \frac{\tilde{C}_2}{(2\varepsilon)^{1/6}}, \quad \frac{\alpha e^{i\pi/4}}{2\sqrt{\pi}} = \frac{\tilde{C}_1}{(2\varepsilon)^{1/6}}$$

Finally, the boundary condition $y(-1) = a$ implies we must have

$$a = \frac{1}{|2 \tanh(-1)|^{1/4}} [\tilde{C}_1 \exp\left\{\frac{iA_0}{\varepsilon}\right\} + \tilde{C}_2 \exp\left\{-\frac{iA_0}{\varepsilon}\right\}]$$

where $A_0 = \int_0^{-1} |2 \tanh t|^{1/2} dt \approx -0.8863$. Then we have four relations which could be solved to determine $C_2, \tilde{C}_1, \tilde{C}_2$.

5.3 WKB applied to boundary value problems

Let us return to the linear boundary value problem that we considered at the very beginning of the course, namely

$$\varepsilon y'' + p(x)y' + q(x)y = 0 \quad x \in (0, 1)$$

with $y(0) = A$ and $y(1) = B$.

Let us restrict ourselves to the case where $p(x) > 0$ for $x \in (0, 1)$. In this case, recall, there is a boundary layer at $x = 0$.

Let us seek a solution of the WKB form

$$y = (y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \cdots) \exp\left\{\frac{\theta(x)}{\varepsilon}\right\}$$

Then

$$y' = (y'_0 + \varepsilon y'_1 + \cdots) \exp\left\{\frac{\theta}{\varepsilon}\right\} + \frac{\theta'}{\varepsilon} (y_0 + \varepsilon y_1 + \cdots) \exp\left\{\frac{\theta}{\varepsilon}\right\}$$

$$y'' = [(y'_0 + \dots) + \frac{\theta'}{\varepsilon}(y'_0 + \dots) + \frac{\theta''}{\varepsilon}(y_0 + \dots) + \frac{\theta'}{\varepsilon}(y'_0 + \dots) + (\frac{\theta'}{\varepsilon})^2(y_0 + \varepsilon y_1 + \dots)] \exp\left\{\frac{\theta}{\varepsilon}\right\}$$

Thus, substitute into the ODE we have

$$\begin{aligned} & \varepsilon\left(\frac{2\theta'}{\varepsilon}y'_0 + \frac{\theta''}{\varepsilon}y_0 + (\frac{\theta'}{\varepsilon})^2y_0 + \frac{(\theta')^2}{\varepsilon}y_1 + \dots\right) \exp\left\{\frac{\theta}{\varepsilon}\right\} \\ & + p(x)(y_0 + \frac{\theta'y_0}{\varepsilon} + \theta'y_1 + \dots) \exp\left\{\frac{\theta}{\varepsilon}\right\} + q(x)(y_0 + \dots) \exp\left\{\frac{\theta}{\varepsilon}\right\} = 0 \end{aligned}$$

This gives at $O(\varepsilon^{-1})$ we have

$$(\theta')^2 y_0 + p(x)\theta' y_0 = 0 \implies (\theta')^2 + p(x)\theta' = 0$$

This has the two possible solutions

$$\theta' = 0 \implies \theta(x) = \text{constant}$$

and without loss of generality we can in fact take $\theta(x) = 0$. The second solution is given by

$$\theta' + p(x) = 0 \implies \theta(x) = \int_0^x -p(t) dt$$

We take the lower limit of integration here to be 0. At $O(1)$,

$$\begin{aligned} y_0\theta'' + 2y'_0\theta' + y_1(\theta')^2 + py'_0 + py_1\theta' + qy_0 &= 0 \\ \implies y_0\theta'' + 2y'_0\theta' + py'_0 + qy_0 &= 0 \end{aligned}$$

Now solve this transport equation for each of the two possible solutions for θ :

(1) For $\theta = 0$ we have

$$py'_0 + qy_0 = 0 \implies y_0 = C_1 \exp\left\{-\int_0^x \frac{q(t)}{p(t)} dt\right\}$$

where C_1 is a constant.

(2) If $\theta = -\int_0^x p(t) dt$ then

$$\theta' = -p(x), \theta'' = -p'(x)$$

Now the transport equation becomes

$$-p'y_0 - 2y'_0p + py_0 + qy_0 = 0 \implies p'y_0 + py'_0 - qy_0 = 0$$

i.e. $(py_0)' - qy_0 = 0$ denote $z = py_0$ so

$$\begin{aligned} z' - \frac{q}{p}z &= 0 \implies z = C_2 \exp\left\{\int_0^x \frac{q(t)}{p(t)} dt\right\} \\ \implies y_0 &= \frac{C_2}{p(x)} \exp\left\{\int_0^x \frac{q(t)}{p(t)} dt\right\} \end{aligned}$$

So to leading order, the WKB solution of the ODE is

$$y_{\text{WKB}}(x) = C_1 \exp\left\{-\int_0^x \frac{q(t)}{p(t)} dt\right\} + \frac{C_2}{p(x)} \exp\left\{\int_0^x \frac{q(t)}{p(t)} dt - \frac{1}{\varepsilon} \int_0^x p(t) dt\right\}$$

Now apply boundary conditions $y(0) = A$ then we have $C_1 + \frac{C_2}{p(0)} = A$ and $y(1) = B \implies$

$$C_1 \exp\left\{-\int_0^1 \frac{q(t)}{p(t)} dt\right\} = B \implies C_1 = B \exp\left\{\int_0^1 \frac{q(t)}{p(t)} dt\right\}$$

there relations determine C_1 and C_2 . Thus,

$$y_{\text{WKB}}(x) = B \exp\left\{\int_x^1 \frac{q(t)}{p(t)} dt\right\} + (A - B \exp\left\{\int_0^1 \frac{q(t)}{p(t)} dt\right\}) \frac{p(0)}{p(x)} \exp\left\{\int_0^x \frac{q(t)}{p(t)} dt - \frac{1}{\varepsilon} \int_0^x p(t) dt\right\}$$

Example. Recall the example

$$\varepsilon y'' + y' + y = 0, \quad x \in (0, 1)$$

with $y(0) = 0$ and $y(1) = e^{-1}$.

We found the composite solution

$$y_c(x; \varepsilon) = e^{-x} - (1+x)e^{-x/\varepsilon} + \varepsilon((1-x)e^{-x} - e^{-x/\varepsilon}) + \dots$$

Now, the above solution found using the WKB method is

$$y_{\text{WKB}} = e^{-x} - e^x e^{-x/\varepsilon}$$

Note that for $x = O(1)$, $e^{-x/\varepsilon}$ is transcendentally small so to leading order, both of these solutions are e^{-x} . And for $x = O(\varepsilon)$, writing $e^x = 1 + x + \dots$ we have

$$y_{\text{WKB}}(x) = e^{-x} - (1+x)e^{-x/\varepsilon}$$

which agrees with y_c .

Example. For $0 < \varepsilon \ll 1$, consider the following boundary value problem for a real-valued function $y = y(x; \varepsilon)$ for $x \in [-1, \infty)$:

$$\varepsilon^2 y'' - \frac{x}{4}(1+x^2)^2 y = 0$$

with $y(-1; \varepsilon) = 1$, $y(x; \varepsilon) \rightarrow 0$ as $x \rightarrow \infty$.

Seek y in the WKB form

$$y = (y_0(x) + \varepsilon y_1(x) + \dots) \exp\left\{\frac{\theta(x)}{\varepsilon}\right\}$$

then

$$\begin{aligned} y' &= (y_0' + \dots + \frac{\theta'}{\varepsilon} y_0 + \theta' y_1 + \dots) \exp\left\{\frac{\theta(x)}{\varepsilon}\right\} \\ y'' &= \left(\frac{\theta'' y_0 + 2y_0' \theta' + \theta'^2 y_1}{\varepsilon} + \frac{\theta'^2 y_0}{\varepsilon^2}\right) \exp\left\{\frac{\theta(x)}{\varepsilon}\right\} \end{aligned}$$

Now, substitute into ODE we then get at $O(1)$,

$$(\theta')^2 y_0 - q(x) y_0 = 0$$

where $q(x) = \frac{x}{4}(1+x^2)^2$.

$$\implies (\theta')^2 = q \implies \theta = \pm \int_0^x (q(t))^{1/2} dt = \pm \frac{1}{2} \int_0^x t^{1/2}(1+t^2) dt = \pm \left(\frac{x^{3/2}}{3} + \frac{x^{7/2}}{7}\right)$$

At $O(\varepsilon)$, we have

$$\begin{aligned} \theta'' y_0 + 2\theta' y_0' + (\theta')^2 y_1 - q y_1 &= 0 \\ \implies \theta'' y_0 + 2\theta' y_0' &= 0 \\ \implies \frac{y_0'}{y_0} &= -\frac{1}{2} \frac{\theta''}{\theta'} \\ \implies y_0 &= C(\theta')^{-1/2} \end{aligned}$$

for some constant C . Thus,

$$y_0 = \frac{C}{q^{1/4}} = \frac{C}{(\frac{x}{4}(1+x^2)^2)^{1/4}}$$

So we have for $x < 0$,

$$y = y_l(x) = \frac{1}{(\frac{|x|}{4}(1+x^2)^2)^{1/4}} [C_1 \exp\left\{\frac{i}{\varepsilon}|x|^{3/2}(\frac{1}{3} + \frac{x^2}{7})\right\} + C_2 \exp\left\{-\frac{i}{\varepsilon}|x|^{3/2}(\frac{1}{3} + \frac{x^2}{7})\right\}]$$

for some constants C_1 and C_2 to be determined. And for $x > 0$,

$$y = y_r(x) = \frac{1}{(\frac{x}{4}(1+x^2)^2)^{1/4}} [C_3 \exp\left\{\frac{1}{\varepsilon}x^{3/2}(\frac{1}{3} + \frac{x^2}{7})\right\} + C_4 \exp\left\{-\frac{1}{\varepsilon}x^{3/2}(\frac{1}{3} + \frac{x^2}{7})\right\}]$$

for some constants C_3 and C_4 to be determined. Note that in order for $y_l(x)$ to be real-valued, we must have $C_1 = \overline{C_2}$. Therefore we must have

$$C_1 \exp\left\{\frac{i}{\varepsilon}(\frac{1}{3} + \frac{1}{7})\right\} + \overline{C_1} \exp\left\{-\frac{i}{\varepsilon}(\frac{1}{3} + \frac{1}{7})\right\} = 1$$

And in order that $y_r(x) \rightarrow 0$ as $x \rightarrow +\infty$, we must have $C_3 = 0$. Now solution local to $x = 0$, let $X = \frac{x}{\delta(\varepsilon)}$ for some $\delta(\varepsilon)$ to be chosen, and $Y(X) = y(x) = y(\delta X)$ then our ODE becomes

$$\frac{\varepsilon^2 Y''}{\delta^2} - q(x) Y = 0$$

Note that $q(x) = q(\delta X) = q'(0)\delta X$ which is the first non-zero term of the Taylor series about 0 in this case $q'(0) = \frac{\delta X}{4}$.

$$\implies \varepsilon^2 Y'' - \frac{\delta^3 XY}{4} = 0$$

Now choose $\frac{\delta^3}{4} = \varepsilon^2$ i.e. $\delta = (2\varepsilon)^{2/3}$ then we have

$$Y'' - XY = 0$$

Hence

$$Y(X) = \alpha Ai(X) + \beta Bi(X)$$

for some constants to be determined. We required

$$\lim_{X \rightarrow +\infty} Y(X) = \lim_{x \downarrow 0} y_r(x)$$

It follows that $\beta = 0$. For $x = \delta X$ we have

$$\begin{aligned} y_r(x) &= \frac{C_4}{(\frac{\delta X}{4} + \dots)^{1/4}} \exp\left\{-\frac{(\delta X)^3/2}{3\varepsilon} + \dots\right\} = \frac{C_4}{(\frac{\varepsilon}{4})^{1/2} X^{1/2}} \exp\left\{-\frac{2}{3} X^{3/2}\right\} \\ \alpha Ai(X) &= \frac{\alpha}{2\sqrt{\pi} X^{1/4}} \exp\left\{-\frac{2}{3} X^{3/2}\right\} \\ \implies \frac{C_4}{(\varepsilon/4)^{1/2}} &= \frac{\alpha}{2\sqrt{\pi}} \end{aligned}$$

We also require

$$\begin{aligned} \lim_{X \rightarrow -\infty} Y(X) &= \lim_{x \uparrow 0} y_l(x) \\ \implies C_4 &= \frac{\alpha \varepsilon^{1/2} e^{-i\pi/4}}{2^{4/3} \sqrt{\pi}} \end{aligned}$$

And

$$C_1 = \frac{e^{-i\pi/4}}{2 \cos\left(\frac{\frac{10}{21\varepsilon}}{1} - \frac{\pi}{4}\right)}$$
$$C_4 = \frac{1}{2 \cos\left(\frac{10}{21\varepsilon} - \frac{\pi}{4}\right)}$$